Past Problems

for Korean Undergraduate Mathematics Competition

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2001.0.1. For $n \times n$ matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_n \end{pmatrix} = \det(\mathbf{A} - \mathbf{BC}).$$

2001.0.2. Find the general solution of

$$y'' - \frac{1}{x}y' - 4x^2 y = 0, \qquad x > 0.$$

Hint. $y = e^{x^2}$ is a solution of the differential equation.

2001.0.3. Let $f\colon [0,\infty) \to [0,\infty)$ satisfy

$$2f(x)f(1) \le f(x)f(1) + 1 \le 2f(2x)$$

for every $x \ge 0$. Prove that f is constant.

2001.0.4. A triangle ABC is inscribed in a circle of radius 1. Let a, b, c be the side lengths opposite the angles A, B, C, respectively. Find the maximum value of

$$a\cos A + b\cos B + c\cos C.$$

2001.0.5. Evaluate the infinite series

$$\sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{2}\right)^{3k+2} \frac{9k+5}{(3k+1)(3k+2)}.$$

2001.0.6. Let $\lambda \ge 0$ and $0 < p < 1$. Suppose f is defined for $x > 0$ and satisfies

$$f(x) = (x - \lambda f(x))^p.$$

Let $g(x) = x^p$. Prove that, for every $x > 0$,

$$\frac{f'(x)}{g'(x)} < \frac{1}{p}.$$

2001.2.1. Let $0 < a, b < 1$ satisfy

$$\int_0^a \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x = \int_b^1 \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x.$$

Find $a^2 + b^2$.

2001.2.2. Let $f\colon [a,b] \to \mathbb{R}$ have a continuous derivative. Evaluate

$$\lim_{\lambda\to\infty} \int_a^b f(x)\left(|\sin(\lambda x)| - \frac{2}{\pi}\right)\mathrm{d}x.$$

2001.2.3. Let $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ be an orthonormal basis of $\mathbb{R}^n$, and let $\mathbf{x} \in \mathbb{R}^n$ be a unit vector. Prove that

$$\sum_{i=1}^{n} \|\mathbf{x} - \mathbf{v}_i\|^2 \geq 2n - 2\sqrt{n}.$$

2001.1.1. Show that there are infinitely many triples $(a, b, c) \in \mathbb{N}^3$ satisfying

$$a\cos\left(\arctan\frac{b}{a}\right) + b\sin\left(\arctan\frac{b}{a}\right) = c.$$

2001.1.2. Evaluate

$$\lim_{n\to\infty} \int_0^{\pi/2} \frac{\sin x}{n^2\left(x - \frac{\pi}{4}\right)^2 + 1}\,\mathrm{d}x.$$

2001.1.3. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n \in \mathbb{R}^2$ satisfy

$$\|\mathbf{v}_i\| \leq 1 \qquad (i = 1, 2, \ldots, n).$$

Prove that there always exist signs

$$\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n \in \{1, -1\}$$

such that

$$\|\varepsilon_1\mathbf{v}_1 + \varepsilon_2\mathbf{v}_2 + \cdots + \varepsilon_n\mathbf{v}_n\| \leq \sqrt{2}.$$

2002.0.1. Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous. Suppose the function $x \mapsto xf(x)$ is differentiable for $x \neq 0$ and

$$\frac{\mathrm{d}}{\mathrm{d}x}(xf(x)) = e^x.$$

Find f.

2002.0.2. Let (a_n) be an arithmetic sequence with first term a and common difference d, and define the 2002×2002 matrix

$$\mathbf{A} \coloneqq \begin{pmatrix} a_1 & a_2 & a_3 & a_4 & \cdots & a_{2002} \\ a_2 & a_1 & a_2 & a_3 & \cdots & a_{2001} \\ a_3 & a_2 & a_1 & a_2 & \cdots & a_{2000} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{2002} & a_{2001} & a_{2000} & a_{1999} & \cdots & a_1 \end{pmatrix}.$$

Express $\det(\mathbf{A})$ in terms of a and d.

2002.0.3. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ be unit vectors in $\mathbb{R}^n$. Prove that

$$|\langle \mathbf{a}, \mathbf{c}\rangle\langle \mathbf{b}, \mathbf{d}\rangle - \langle \mathbf{a}, \mathbf{d}\rangle\langle \mathbf{b}, \mathbf{c}\rangle| \leq 1.$$

2002.0.4. Let $n \in \mathbb{N}$, and let $a_1, \ldots, a_n, x_1, \ldots, x_n$ be positive real numbers satisfying

(i) $a_1 \leq a_2 \leq \cdots \leq a_n$;

(ii) $x_k \geq x_n$ for $k = 1, 2, \ldots, n-1$.

Prove that

$$(a_1 + x_1a_2)(a_2 + x_2a_3)\cdots(a_{n-1} + x_{n-1}a_n)(a_n + x_na_1) \geq a_1a_2\cdots a_n(x_1+1)(x_2+1)\cdots(x_n+1).$$

2002.0.5. For $\alpha \in \mathbb{C}$, define the $n \times n$ matrix $\mathbf{A}_n = (a_{ij})$ by

$$a_{ij} = \begin{cases} 1, & i = j, \\ \alpha, & i - j = -1, \\ \overline{\alpha}, & i - j = 1, \\ 0, & |i - j| \geq 2. \end{cases}$$

Prove that $\mathbf{A}_n$ is positive semidefinite for every $n = 1, 2, \ldots$ if and only if

$$|\alpha| \leq \frac{1}{2}.$$

2002.0.6. Solve the system of differential equations

$$x''(t) + 2y'(t) - x(t) = 0, \qquad y''(t) - 2x'(t) - y(t) = 0$$

subject to

$$x(0) = y(0) = 0, \qquad x'(0) = a > 0, \qquad y'(0) = 0.$$

2002.2.1. For $(x, y) \neq (0, 0)$, define

$$f(x, y) = \frac{x^2 - xy + y^2}{2x^2 + y^2}.$$

Find the minimum value of f.

2002.2.2. Let $f, g, h\colon [-1, 1] \to \mathbb{R}$ be continuous. Suppose that f is odd, g is even, and

$$\int_{-1}^{1} f(x)^2 \, \mathrm{d}x = \int_{-1}^{1} g(x)^2 \, \mathrm{d}x = 1.$$

Prove that

$$\left(\int_{-1}^{1} f(x)h(x) \, \mathrm{d}x\right)^2 + \left(\int_{-1}^{1} g(x)h(x) \, \mathrm{d}x\right)^2 \leq \int_{-1}^{1} h(x)^2 \, \mathrm{d}x.$$

2002.2.3. Let $a > b > 0$. Evaluate

$$\int_0^\infty \frac{\arctan(ax) - \arctan(bx)}{x} \, \mathrm{d}x.$$

2002.1.1. Let $f\colon [a,b] \to \mathbb{R}$ have a continuous derivative, with $f(a) = 0$. Suppose that

$$\left|\frac{f'(x)}{2003}\right| \leq \left|\frac{f(x)}{2002}\right| \qquad (x \in [a,b]).$$

Prove that $f(x) \equiv 0$ on $[a,b]$.

2002.1.2. Let $n \in \mathbb{N}$ with $n > 1$. Prove that

$$\frac{\big((n-1)!\big)^{1/(n-1)}}{(n!)^{1/n}} < \frac{(n!)^{1/n}}{\big((n+1)!\big)^{1/(n+1)}}.$$

It is given that

$$4 > e^{4-2\sqrt{2}}.$$

2002.1.3. Let

$$\alpha = 0.0110101000101000101\ldots,$$

where the nth digit after the decimal point is 1 if n is prime and 0 otherwise. Prove that α is irrational.

2003.0.1. Find the value of the natural number

$$\sum_{k=1}^{\infty} \binom{10^{10}}{\lfloor \log_{11} k \rfloor}.$$

2003.0.2. Four points A, B, C, D are given in clockwise order on a fixed circle, and the quadrilateral $ABCD$ contains the center of the circle. Let P, Q, R, S be the incenters of the triangles ABC, BCD, CDA, and DAB, respectively. Prove that $PQRS$ is a rectangle.

2003.0.3. Let $a, b \in \mathbb{N}$ and let p, q be primes with $p < 2003$ and $q < 2003$. Find all ordered pairs

$$(p^a, q^b)$$

satisfying

$$p^a - q^b = 1.$$

2003.0.4. For each $k = 1, 2, \ldots, 2002$, let x_k be a real number satisfying

$$k - 1 \leq x_k \leq k.$$

Define

$$B_k = \left\{ \frac{2}{k} \sum_{i=1}^{k} |x - x_i| \,\middle|\, x \in \mathbb{R} \right\}$$

and

$$A = \left\{ \sum_{k=1}^{2002} y_k \,\middle|\, y_k \in B_k,\ k = 1, 2, \ldots, 2002 \right\}.$$

Prove that

$$\binom{2003}{2} \in A.$$

2003.0.5. Prove that the real number $\sin 1$ is irrational.

2003.0.6. Let $f\colon [0,3] \to \mathbb{R}$ be continuous and suppose that

$$\int_0^3 x^k f(x)\ \mathrm{d}x = 0 \qquad (k = 0, 1, 2, \ldots, n-1)$$

and

$$\int_0^3 x^n f(x)\ \mathrm{d}x = 3.$$

Prove that there exists an interval $A \subseteq [0,3]$ such that

$$|f(x)| \geq \left(\frac{2}{3}\right)^n (n+1)$$

for every $x \in A$.

2003.2.1. Let (x_n) be a sequence satisfying

$$0 \leq x_{n+1} \leq x_n$$

for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} x_n$$

converges. Determine

$$\lim_{n\to\infty} n x_n.$$

2003.2.2. Let $\mathbf{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ satisfy

$$x_1 + x_2 + \cdots + x_n = 0.$$

Prove that

$$\max_{1\leq j\leq n} |x_j| \leq \left(\frac{n-1}{n}\right)^{1/2} \|\mathbf{x}\|_2,$$

where

$$\|\mathbf{x}\|_2 = (x_1^2 + x_2^2 + \cdots + x_n^2)^{1/2}.$$

2003.2.3. Let f be the 2π-periodic function defined by

$$f(x) = \begin{cases} \pi + x, & -\pi < x \leq 0, \\ \pi - x, & 0 < x \leq \pi. \end{cases}$$

Using the Fourier series of f, prove that

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \cdots + \frac{1}{(2n-1)^2} + \cdots.$$

2003.1.1. Let (a_n) be a sequence with $a_n \geq 0$ for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} a_n$$

converges. Prove that, for every $p > \frac{1}{2}$, the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$$

converges. Also give an example showing that, when $p = \frac{1}{2}$, the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$$

may diverge.

2003.1.2. Let $p \geq 1$ be a real number, and let $x_1, x_2, \ldots, x_n > 0$ satisfy

$$x_1 + x_2 + \cdots + x_n = 1.$$

Prove that

$$\sum_{j=1}^{n} \left(x_j + \frac{1}{x_j} \right)^p \geq \frac{(1+n^2)^p}{n^{p-1}}.$$

2003.1.3. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix whose entries are all 0 or 1. Suppose that, for some $t \in \mathbb{N}$,

$$\mathbf{A}\mathbf{A}^{\mathsf{T}} = \begin{pmatrix} t+1 & 1 & \cdots & 1 & 1 \\ 1 & t+1 & \cdots & 1 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \cdots & t+1 & 1 \\ 1 & 1 & \cdots & 1 & t+1 \end{pmatrix} = t\mathbf{I}_n + \mathbf{J},$$

where $\mathbf{J}$ is the $n \times n$ all-ones matrix. Express $\det(\mathbf{A})$ as a function of t.

2004.0.1. Let $f : \mathbb{R} \to \mathbb{R}$ satisfy

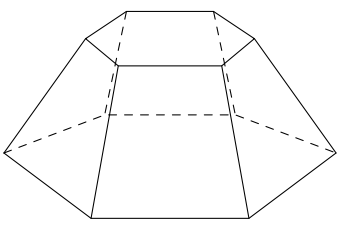
$$f(x+y) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$. Prove the following.

(a) If f is continuous at 0, then f is continuous at every $x \in \mathbb{R}$.

(b) If f is continuous on $\mathbb{R}$ and $f(1) = c$, then $f(x) = cx$ for every $x \in \mathbb{R}$.

2004.0.2. A frustum of a pyramid has parallel upper and lower bases and uniform density. Find the ratio of the distance from its center of mass to the lower base to the distance from its center of mass to the upper base. The similarity ratio of the upper base to the lower base is r, where $r < 1$.



2004.0.3. A convex quadrilateral $ABCD$ has side lengths a, b, c, d, in this order starting from AB. Show that if its area is maximal among all such quadrilaterals, then $ABCD$ is cyclic.

2004.0.4. Let $(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ be an orthonormal basis of $\mathbb{R}^n$. Find the orthogonal complement of

$$\operatorname{span}\{i\mathbf{v}_i - j\mathbf{v}_j \mid 1 \leq i < j \leq n\}.$$

2004.0.5. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function whose second derivative f'' exists and is bounded on $[0, 1]$. Let (a_n) be a sequence such that $0 < a_n < 1$ for every $n \in \mathbb{N}$, the series

$$\sum_{n=1}^{\infty} a_n$$

diverges, and the series

$$\sum_{n=1}^{\infty} a_n^2$$

converges. If

$$\sum_{n=1}^{\infty} f(a_n)$$

converges, prove that

$$\sum_{n=1}^{\infty} |f(a_n)|$$

also converges.

2004.0.6. Suppose a plane curve $(x(t), y(t))$ of period T satisfies

$$\frac{\mathrm{d}x}{\mathrm{d}t} = ax - bxy, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = -cy + dxy, \qquad x(0) = y(0) = 1,$$

where $a, b, c, d > 0$. Find

$$\overline{x} \coloneqq \frac{1}{T}\int_0^T x(t)\ \mathrm{d}t, \qquad \overline{y} \coloneqq \frac{1}{T}\int_0^T y(t)\ \mathrm{d}t.$$

2004.0.7. Let B be any 30-element subset of $\{1, 2, \ldots, 39, 40\}$. Prove that there exists a subset $A \subseteq B$ such that

(i) $|A| \geq 10$;

(ii) $x + y \notin A$ for all $x, y \in A$.

2004.2.1. Let $f\colon [a, c] \to (0, \infty)$ be continuous on $[a, c]$ and differentiable on (a, c). Let

$$P = (x_0, f(x_0))$$

be a point on the graph of f. Let α and γ be the angles made with the x-axis by the lines joining P to $(a, 0)$ and $(c, 0)$, respectively, and let β be the angle made with the x-axis by the tangent line to the graph at P. Show that there exists $x_0 \in (a, c)$ such that

$$\tan\alpha + \tan\beta + \tan\gamma = 0.$$

If the tangent line does not meet the x-axis, take $\beta = 0$.

2004.2.2. Evaluate

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} \, \mathrm{d}x.$$

2004.1.1. Let A be the least positive integer of the form

$$33^k - 7^\ell, \qquad k, \ell \in \mathbb{N},$$

and let B be the least positive integer of the form

$$7^m - 33^n, \qquad m, n \in \mathbb{N}.$$

Find $A + B$.

2004.1.2. Let $\mathbf{v} \in \mathbb{R}^n$ be a unit vector, and let $\mathbf{P_v}$ denote the orthogonal projection onto the line span($\mathbf{v}$), so that

$$\mathbf{P_v x} = (\mathbf{x} \cdot \mathbf{v})\mathbf{v}$$

for every $\mathbf{x} \in \mathbb{R}^n$. Let $\mathbf{A}$ be an $n \times n$ positive definite symmetric matrix. Find the maximum value of λ such that

$$\mathbf{A} - \lambda \mathbf{P_v} \succeq 0.$$

2005.0.1. Let

$$\mathbf{A} = \frac{1}{2} \begin{pmatrix} 0 & \sqrt{2} \\ -\sqrt{2} & 0 \end{pmatrix}.$$

Find the matrix

$$e^{\mathbf{A}} = \sum_{n=0}^{\infty} \frac{\mathbf{A}^n}{n!}.$$

2005.0.2. Solve the differential equation

$$y'(t) + 2ty(t) = ty(t)^2, \qquad y(1) = 1.$$

2005.0.3. Let $p \geq 3$ be an arbitrary prime, and let

$$\mathbb{F}_p = \{0, 1, \ldots, p-1\}.$$

Show that the set

$$S = \{(x, y) \in \mathbb{F}_p^2 \mid x^2 + y^2 \equiv -1 \pmod{p}\}$$

is nonempty.

2005.0.4. Let $f \colon \mathbb{R} \to \mathbb{R}$ be differentiable and suppose that

$$|f(n)| \leq \frac{1}{(1+n)^{5/4}}$$

for every nonnegative integer n, and that

$$|f'(x)| \leq e^{-x}$$

for every $x \in \mathbb{R}$. Prove that

$$\int_0^\infty |f(x)| \, \mathrm{d}x < \infty.$$

2005.0.5. Let S_n be the set of all bijections from $\{1, 2, \ldots, n\}$ to itself. For $\sigma \in S_n$, let $f(\sigma)$ denote the number of fixed points of σ, that is,

$$f(\sigma) = |\{x \in \{1, 2, \ldots, n\} \mid \sigma(x) = x\}|.$$

Find

$$\sum_{\sigma \in S_n} f(\sigma).$$

2005.0.6. A population consists of smokers and nonsmokers. Each year, 5% of the nonsmokers begin smoking and 10% of the smokers quit smoking. Prove that, after a sufficiently long time, the ratio of smokers to nonsmokers converges to a constant, and find that ratio.

2005.0.7. Words of length n are formed using the three letters A, B, C, with the restriction that A may not occur twice consecutively. Express the number of such words as a function of n.

2005.2.1. Let X be the set of all invertible 2×2 matrices and Y the set of all invertible 3×3 matrices. Define $\Phi \colon X \to Y$ by

$$\Phi\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \begin{pmatrix} a^2 & 2ab & b^2 \\ ac & ad + bc & bd \\ c^2 & 2cd & d^2 \end{pmatrix}.$$

Suppose $\mathbf{A} \in X$ has eigenvalues $1, x$ and $\mathbf{B} \in X$ has eigenvalues $1 + x, 1 - x$. Find

$$\det\big(\Phi(\mathbf{AB})\big).$$

2005.2.2. Let $f \colon (0, \infty) \to \mathbb{R}$ be twice differentiable and suppose that

$$|f(x)| \leq 1, \qquad |f''(x)| \leq 1$$

for every $x \in (0, \infty)$. Prove that

$$|f'(x)| \leq 2$$

for every $x \in (0, \infty)$.

2005.1.1. Let $f \colon [0, 1] \to \mathbb{R}$ be differentiable, with $f(0) = 0$ and $f(1) = 1$. Prove that, for every $n \in \mathbb{N}$, there exist distinct points $x_1, x_2, \ldots, x_n \in [0, 1]$ such that

$$\sum_{k=1}^{n} \frac{1}{f'(x_k)} = n.$$

2005.1.2. Let (a_n) and (b_n) be sequences satisfying $a_n > 0$ and $b_n > 0$ for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} a_n = 3, \qquad \sum_{n=1}^{\infty} b_n = 4.$$

Find the supremum and infimum of all possible values of

$$\sum_{n=1}^{\infty} \frac{a_n^2}{a_n + b_n}.$$

2006.0.1. Let $f(x) = \arctan(2x)$. Find $f^{(99)}(0)$.

2006.0.2. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous periodic function with period $2T$. Prove that there exists a suitable $x_0 \in \mathbb{R}$ such that

$$f(x_0 + T) = f(x_0).$$

2006.0.3. Let $\mathbf{A}$ be an $n \times n$ matrix such that

$$\operatorname{tr}(\mathbf{XYA}) = \operatorname{tr}(\mathbf{YXA})$$

for every pair of $n \times n$ matrices $\mathbf{X}, \mathbf{Y}$. Prove that $\mathbf{A}$ is a scalar multiple of the identity matrix.

2006.0.4. Let (a_n) be a sequence of positive real numbers, and suppose that the limit

$$\lim_{n\to\infty} n\left(\frac{a_n}{a_{n+1}} - 1\right)$$

exists. Prove that if this limit is greater than 1, then the series

$$\sum_{n=1}^{\infty} a_n$$

converges.

2006.0.5. Let $f\colon (0,\infty) \to \mathbb{R}$ be differentiable, suppose that f' is continuous, and assume that

$$\lim_{x\to\infty} \big(f(x) + f'(x)\big) = 0.$$

Prove that

$$\lim_{x\to\infty} f(x) = 0.$$

2006.0.6. Let $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ be a unit vector, and let $\mathbf{X} = (x_{ij})$ be the $n \times n$ matrix defined by

$$x_{ij} = \delta_{ij} + a_i a_j.$$

Compute $\det(\mathbf{X})$.

2006.0.7. Let $h\colon \mathbb{R} \to \mathbb{R}$ satisfy

$$\int_0^{\infty} h(t)\ \mathrm{d}t = 1.$$

Evaluate

$$\iint_{\mathbb{R}^2} h(x^2 + 2xy + 5y^2)\ \mathrm{d}A.$$

2006.2.1. Evaluate

$$\int_{-\pi}^{\pi} \ln\left|\sin\frac{\theta}{2}\right|\ \mathrm{d}\theta.$$

2006.2.2. Evaluate the series

$$1 - \frac{1}{5} + \frac{1}{7} - \frac{1}{11} + \cdots + \frac{1}{6n-5} - \frac{1}{6n-1} + \cdots.$$

2006.1.1. Prove that the improper integral

$$\int_0^\infty \sin(x^2)\ \mathrm{d}x$$

converges, whereas

$$\int_0^\infty |\sin(x^2)|\ \mathrm{d}x$$

diverges.

2006.1.2. Consider the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots.$$

Let $\sum a_k$ be a series obtained by rearranging its terms subject to the rule that the relative order among terms of the same sign is not changed. For example,

$$1 + \frac{1}{3} - \frac{1}{2} - \frac{1}{4} + \cdots$$

is allowed, whereas

$$1 - \frac{1}{4} + \frac{1}{3} - \frac{1}{2} + \cdots$$

is not. Among $a_1, a_2, \ldots, a_n$, let p_n be the number of positive terms and q_n the number of negative terms.

(a) Prove that $\sum a_k$ converges if and only if the limit

$$\lim_{n\to\infty} \frac{p_n}{q_n}$$

exists.

(b) If this limit is α, prove that

$$\sum a_k = \ln 2 + \frac{1}{2}\ln \alpha.$$

2007.0.1. Find the area of the quadrilateral whose vertices are

$$(0,0,0), \quad (1,2,3), \quad (3,1,2), \quad (7,4,7).$$

2007.0.2. Let

$$f(x) = \frac{e^x}{x}, \qquad 1 \le x \le 2,$$

and let g be the inverse function of f. Evaluate

$$\int_e^{e^2/2} [g(x)]^2\ \mathrm{d}x.$$

2007.0.3. Three unit vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$ determine a parallelepiped of volume $\frac{1}{2}$. Let θ_{ij} be the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$, and let $\mathbf{A}$ be the 3×3 matrix whose (i,j)-entry is $\cos\theta_{ij}$. Compute $\det(\mathbf{A})$.

2007.2.1. Evaluate

$$\int_0^1 \int_0^x (3 - x - y)\ \mathrm{d}y\ \mathrm{d}x.$$

2007.2.2. Let $C(t) = (\cos t, \sin t)$ for $0 \le t \le 2\pi$. Evaluate the line integral

$$\int_C \frac{y\ \mathrm{d}x - x\ \mathrm{d}y}{x^2 + y^2}.$$

2007.2.3. Find the functions $x(t)$ and $y(t)$ satisfying

$$x'(t) = -y(t), \qquad y'(t) = x(t), \qquad (x(0), y(0)) = (1, 1).$$

2007.2.4. Find the relation among a, b, c that is necessary and sufficient for the system

$$\begin{aligned} x_1 - 2x_2 - x_3 + 3x_4 &= a, \\ 2x_1 + 4x_2 + 6x_3 - 2x_4 &= b, \\ x_1 + x_3 + x_4 &= c \end{aligned}$$

to have a solution.

2007.2.5. Write the matrix

$$\begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{pmatrix}$$

as $\mathbf{LU}$, where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.

2007.2.6. Find the volume of the region in $\mathbb{R}^3$ described by

$$x = (p - q)\cos t, \qquad y = (p - q)\sin t, \qquad z = p + q,$$

where

$$0 \le t \le 2\pi, \qquad 1 - \varepsilon \le p \le 1 + \varepsilon, \qquad -\delta \le q \le \delta, \qquad \delta < 1 - \varepsilon.$$

2007.2.7. Find the constants a, b that minimize

$$\int_0^\pi [\sin x - (ax + b)]^2\ \mathrm{d}x.$$

2007.2.8. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$ be linearly independent vectors satisfying

$$\|\mathbf{v}_i\| = 1, \qquad \mathbf{v}_i \cdot \mathbf{v}_j = -\frac{1}{3} \quad (i \ne j).$$

For vectors $\mathbf{x} \in \mathbb{R}^3$ with $\|\mathbf{x}\| = 1$, find the maximum and minimum values of

$$\sum_{i=1}^{3} \|\mathbf{x} - \mathbf{v}_i\|^2.$$

2007.2.9. For real y, evaluate

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi iyx}}{\cosh(\pi x)}\,\mathrm{d}x.$$

2007.2.10. Let $p > 0$ be real. Evaluate

$$\lim_{n\to\infty} \frac{1}{n^2} \sum_{k=1}^{n} \left(1^p + 2^p + 3^p + \cdots + k^p\right)^{1/(p+1)}.$$

2007.1.1. Let $A = \{1, 2, 3, \ldots, n\}$. Suppose k distinct subsets of A, each containing exactly γ elements, are selected. Prove that if every element of A is to belong to at least p of the selected subsets, then

$$k \geq \frac{np}{\gamma}.$$

2007.1.2. Let $M_{2\times 2}$ be the vector space of 2×2 matrices, and let $\mathbf{T} \in M_{2\times 2}$ be invertible. Compute the determinant of the linear map

$$\Phi\colon M_{2\times 2} \to M_{2\times 2}, \qquad \Phi(\mathbf{A}) = \mathbf{TAT}^{-1}.$$

2007.1.3. In triangle ABC, let P be a point on the side BC. Put

$$BP = x, \qquad BC = l,$$

let the altitude of the triangle be h, and let

$$\angle APC = \theta(x).$$

Prove that

$$\angle A = \frac{1}{h}\int_0^l \sin^2\theta(x)\,\mathrm{d}x.$$

2007.1.4. Let $\mathbf{A}$ and $\mathbf{B}$ be real matrices. Suppose there exists a complex matrix $\mathbf{S}$ such that

$$\mathbf{A} = \mathbf{S}^{-1}\mathbf{BS}.$$

Prove that there exists a real matrix $\mathbf{R}$ such that

$$\mathbf{A} = \mathbf{R}^{-1}\mathbf{BR}.$$

2007.1.5. Let $a_1 > 0$ and

$$a_{n+1} = \ln(1 + a_n).$$

Prove that

$$\lim_{n\to\infty} na_n = 2.$$

2007.1.6. Let (a_n) be a sequence such that

$$\sum_{n=1}^{\infty} a_n = 1.$$

Prove that

$$\lim_{n\to\infty} \sum_{k=1}^{n} a_k \left(1 - \frac{k^2}{n^2}\right) = 1.$$

2007.1.7. Let $F\colon \mathbb{R}^2 \to \mathbb{R}^2$ satisfy the following condition: for every triangle Δ of area 1, $F(\Delta)$ is also a triangle of area 1. Prove that for every straight line l, $F(l)$ is also a straight line.

2008.0.1. Evaluate

$$\int_0^\infty \frac{e^{-x} - e^{-2x}}{x}\,\mathrm{d}x.$$

2008.0.2. Let $\alpha > 1$. Prove that the improper integral

$$\int_1^\infty \cos(x^\alpha)\,\mathrm{d}x$$

converges.

2008.0.3. For every $n = 0, 1, 2, 3, \ldots$, let a_n be either 0 or 1. Let P and Q be polynomials with $\deg P < \deg Q$, and suppose that for $|x| < 1$,

$$\sum_{n=0}^{\infty} a_n x^n = \frac{P(x)}{Q(x)}.$$

Prove that there exists a positive integer p such that

$$a_{n+p} = a_n$$

for every $n \geq 0$.

2008.2.1. Find the volume of the tetrahedron whose vertices are

$$(0,1,0), \quad (1,2,1), \quad (1,3,3), \quad (3,1,2).$$

2008.2.2. Evaluate

$$\int_0^{\sqrt{\pi/2}} \int_0^{\sqrt{\pi/2}} \int_0^{\sqrt{\pi/2}} xyz\cos(x^2+y^2+z^2)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.$$

2008.2.3. Find one matrix $\mathbf{A}$ satisfying

$$\mathbf{A}^2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

2008.2.4. Find a function $f(x,y,z)$ satisfying

$$\nabla f(x,y,z) = \left(ye^{xy+yz}, (x+z)e^{xy+yz}, ye^{xy+yz}\right).$$

2008.2.5. Find the maximum value of

$$f(x,y) = 3x^2 + 4xy + y^2$$

on the set

$$\{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

2008.2.6. Find all 3×1 column vectors $\mathbf{u}$ and $\mathbf{v}$ satisfying

$$\begin{pmatrix} 5 & 2 & 3 \\ 15 & 6 & 9 \\ 10 & 4 & 6 \end{pmatrix} = \mathbf{u}\mathbf{v}^{\mathsf{T}}.$$

2008.2.7. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix satisfying

$$\sum_{i=1}^{n} a_{ij} = 0$$

for every $j = 1, 2, \ldots, n$. Prove that $\mathbf{A}$ is not invertible.

2008.2.8. Let (ϕ_n) be a sequence of functions defined on $[0, 1]$ satisfying

(a) $\phi_0(t) = 0$;

(b)

$$\phi_n(t) = 1 + 2\sin\frac{t}{2} - \frac{1}{2}\int_0^t \phi_{n-1}(s)\ \mathrm{d}s, \qquad n = 1, 2, \ldots.$$

Suppose that $\phi_n \to \phi$ on $[0, 1]$ and that ϕ is differentiable. Find $\phi(t)$.

2008.2.9. Find all polynomial functions $u(x, y)$ satisfying the following conditions.

(a) $u_{xx} + u_{yy} = 0$.

(b) For every $x \in \mathbb{R}$,

$$u(x, 0) = u(x, \sqrt{3}\,x) = 0.$$

(c) For every $x > 0$ and $0 < y < \sqrt{3}\,x$,

$$u(x, y) > 0.$$

(d) There exists a positive integer n such that, for every $t \in \mathbb{R}$,

$$u(tx, ty) = t^n u(x, y).$$

2008.2.10. For the four data points

$$(0, 0), \quad (1, 1), \quad (3, 2), \quad (4, 5),$$

find the line $y = ax + b$ that minimizes the sum of squared errors

$$E^2 = \sum_{i=1}^{4} (y_i - ax_i - b)^2.$$

2008.1.1. Find the volume of the tetrahedron whose vertices are

$$(0, 1, 0), \quad (1, 2, 1), \quad (1, 3, 3), \quad (3, 2, 1).$$

2008.1.2. Let $0 < a, b, c, d < 1$. Prove the inequality

$$\begin{aligned}\sqrt{(1-ab)(1-cd)} \geq \sqrt{ac(1-b)(1-d)} + \sqrt{bd(1-a)(1-c)} \\ + \sqrt{(1-a)(1-b)(1-c)(1-d)}.\end{aligned}$$

2008.1.3. Evaluate

$$\sin\frac{\pi}{10}\sin\frac{2\pi}{10}\cdots\sin\frac{9\pi}{10}.$$

2008.1.4. Let $\mathbf{A}$ be a 3×3 matrix all of whose entries are positive, and suppose that $\mathbf{A}^{-1}$ exists. If exactly six entries of $\mathbf{A}^{-1}$ are positive, prove that exactly three entries of $\mathbf{A}^{-1}$ are negative.

2008.1.5. Let (ϕ_n) be a sequence of functions defined on $[0,1]$ satisfying

(a) $\phi_0(t)=0$;

(b)

$$\phi_n(t)=1+2\sin\frac{t}{2}-\frac{1}{2}\int_0^t \phi_{n-1}(s)\,\mathrm{d}s, \qquad n=1,2,\ldots.$$

Show that (ϕ_n) converges on $[0,1]$, and determine

$$\phi=\lim_{n\to\infty}\phi_n.$$

2008.1.6. Let $p_i, q_i>0$ and suppose that

$$\sum_{i=1}^n p_i=\sum_{i=1}^n q_i=1.$$

Prove that

$$\frac{1}{2}\sum_{i=1}^n (p_i-q_i)^2\le\sum_{i=1}^n p_i\ln\left(\frac{p_i}{q_i}\right).$$

2008.1.7. Let (x_n) be a sequence satisfying

$$x_m>0, \qquad x_{m+n}\le x_m+x_n, \qquad m,n=1,2,3,\ldots.$$

Prove that the sequence (x_n/n) converges.

2009.0.1. Let $\mathbf{x},\mathbf{y}\in\mathbb{R}^n$ be nonzero column vectors, and suppose that

$$\mathbf{A}=\mathbf{x}\mathbf{y}^{\mathsf{T}}$$

is symmetric. Prove that there exists a vector $\mathbf{u}\in\mathbb{R}^n$ such that

$$\mathbf{A}=\mathbf{u}\mathbf{u}^{\mathsf{T}} \qquad \text{or} \qquad \mathbf{A}=-\mathbf{u}\mathbf{u}^{\mathsf{T}}.$$

2009.0.2. Find all functions $f\colon\mathbb{R}\to\mathbb{R}$ satisfying the following conditions.

(i) $f(x)>0$ for every $x\in\mathbb{R}$.

(ii)

$$f'(x)-6f(x)f'(x)-f'''(x)=0$$

for every $x\in\mathbb{R}$.

(iii)

$$\lim_{x\to\infty}f(x)=\lim_{x\to\infty}f'(x)=\lim_{x\to\infty}f''(x)=0.$$

2009.0.3. Evaluate

$$\int_{-\pi}^{\pi} \frac{x^2}{1+\sin x+\sqrt{1+\sin^2 x}}\,\mathrm{d}x.$$

2009.2.1. Find the length of the curve

$$y=\frac{1}{2}x^2, \qquad -1\le x\le 1.$$

2009.2.2. Find the area of the region in the plane enclosed by

$$x^{2/3}+y^{2/3}=1.$$

2009.2.3. Let

$$\mathbf{A}=\begin{pmatrix}-3 & 1 & 4\\ -4 & 2 & 4\\ -1 & 1 & 2\end{pmatrix}.$$

Find the sum of all entries of $\mathbf{A}^{2009}$.

2009.2.4. Prove that, for every positive integer n,

$$\left(\frac{2}{3}\right)^{2n}+\left(\frac{2}{5}\right)^{2n}+\left(\frac{2}{7}\right)^{2n}+\cdots<\frac{1}{n}.$$

2009.2.5. Let $f\colon[-\pi,\pi]\to\mathbb{R}$ be integrable and suppose that

$$\int_{-\pi}^{\pi} f(x)^2\,\mathrm{d}x<\infty.$$

For each nonnegative integer k, define

$$a_k=\int_{-\pi}^{\pi} f(x)\cos(kx)\,\mathrm{d}x, \qquad b_k=\int_{-\pi}^{\pi} f(x)\sin(kx)\,\mathrm{d}x.$$

Prove that

$$\lim_{k\to\infty} a_k=\lim_{k\to\infty} b_k=0.$$

2009.1.1. Evaluate

$$\int_0^1\int_y^1 \frac{\cos(\ln(x^2+1))}{x^2+1}\,\mathrm{d}x\,\mathrm{d}y.$$

2009.1.2. Let $p\ge 5$ be prime. Express

$$\sum_{k=1}^{p^2}\frac{1}{k}-\frac{1}{p}\sum_{k=1}^{p}\frac{1}{k}$$

as a reduced fraction. Prove that its numerator is divisible by p^4.

2009.1.3. Let f be twice differentiable on $[0,2]$ and suppose that

$$f''(x) \geq 0 \qquad (0 \leq x \leq 2).$$

Prove that

$$\int_0^2 f(x)\ \mathrm{d}x \geq 2f(1).$$

2009.1.4. For a real number a and subsets $A, B \subseteq \mathbb{R}$, define

$$aA = \{ax \mid x \in A\}, \qquad A + B = \{a + b \mid a \in A,\ b \in B\}.$$

For each positive integer n, define

$$G_0 = [0,1) = \{x \mid 0 \leq x < 1\}, \qquad G_n = \frac{1}{5}(\{0,2,4\} + G_{n-1}).$$

Prove that the set

$$G = \bigcap_{n=1}^{\infty} G_n$$

is uncountable.

2009.1.5. Let

$$g(x) = \frac{1}{3}x^2 + \frac{2}{3}x.$$

Suppose that a continuous function $f\colon [0,1] \to [0,1]$ satisfies

$$|f(x) - f(y)| \leq g(|x - y|)$$

for all $x, y \in [0,1]$. Prove that the set

$$A = \{c \in [0,1] \mid f(c) = g(c)\}$$

either consists of a single point or is a closed interval.

2010.2.1. Evaluate

$$\int_0^{\infty} \frac{1}{x^3 + 1}\ \mathrm{d}x.$$

2010.2.2. Find the surface area obtained by rotating the curve

$$y = \frac{x^2}{2}, \qquad 0 \leq x \leq 1,$$

about the x-axis.

2010.2.3. Find the general solution on $x > 1$ of the differential equation

$$(x-1)y'' - xy' + y = 0.$$

2010.2.4. Let $n \geq 3$. Solve the system

$$\begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ 2 & 3 & 4 & \cdots & 1 \\ 3 & 4 & 5 & \cdots & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n & 1 & 2 & \cdots & n-1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} n \\ n-1 \\ n-2 \\ \vdots \\ 1 \end{pmatrix}.$$

2010.2.5. Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices satisfying

$$\mathbf{A} + \mathbf{B} = \mathbf{AB}.$$

Prove that

$$\mathbf{AB} = \mathbf{BA}.$$

2010.2.6. Let $y = f(x)$ be a continuous function having a local extremum at $x = 1$. Suppose that, in a neighborhood of $x = 1$,

$$y^3 + 3xy^2 + x^3y = 1.$$

Find the second-degree Taylor polynomial of f at $x = 1$, and determine whether $f(1)$ is a local maximum or a local minimum.

2010.2.7.

(a) Find a cubic polynomial $f(x)$ with integer coefficients such that

$$f\left(\cos\frac{\pi}{7}\right) = 0.$$

(b) Prove that

$$4\sin\frac{2\pi}{7} - \tan\frac{\pi}{7} = \sqrt{7}.$$

2010.2.8. For $a \in \mathbb{R}$, evaluate

$$\int_0^\infty e^{-x^2}\cos(ax)\ \mathrm{d}x.$$

2010.1.1. Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous and satisfy

$$\sum_{k=0}^{2009} f(x+k) = x^{2009}$$

for every $x \in \mathbb{R}$. Evaluate

$$\int_0^{2010} f(x)\ \mathrm{d}x.$$

2010.1.2. Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Find the minimal polynomial of $\mathbf{A}$, and prove that there is no matrix $\mathbf{B}$ satisfying

$$\mathbf{B}^2 = \mathbf{A}.$$

2010.1.3. Let

$$f(x) = x^m + a_{m-1}x^{m-1} + a_{m-2}x^{m-2} + \cdots + a_1x + a_0$$

divide $x^n - 1$, and let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{m-1} \end{pmatrix}.$$

Prove that

$$\det\Big(x^{n-1}\mathbf{I}_m + x^{n-2}\mathbf{A} + x^{n-3}\mathbf{A}^2 + \cdots + x^2\mathbf{A}^{n-3} + x\mathbf{A}^{n-2} + \mathbf{A}^{n-1}\Big) = \frac{(x^n-1)^m}{f(x)}.$$

2010.1.4. Let p be a prime such that $p-1$ is not divisible by 7. Prove that there are infinitely many positive integers n for which

$$p \mid n^7 - 2010.$$

2010.1.5. Find all real numbers a for which the improper integral

$$\int_0^\infty \frac{\sin^2 x}{x^a}\,\mathrm{d}x$$

converges.

2010.1.6. Let $f\colon \mathbb{R} \to \mathbb{R}$ have a continuous derivative. For $x \neq y$, define

$$g(x,y) = \frac{f(x)-f(y)}{x-y}.$$

Prove that $g(x,x)$ can be defined so that g is continuous on $\mathbb{R}^2$.

2010.1.7. Evaluate

$$\int_{1/2}^2 \frac{\arctan x}{x^2-x+1}\,\mathrm{d}x.$$

2010.1.8. Find positive real numbers a and r satisfying

$$\lim_{n\to\infty} n^r \cdot \frac{1}{2}\cdot\frac{3}{4}\cdots\frac{2n-1}{2n} = a.$$

2011.0.1. Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \begin{cases} i^2-1, & i = j, \\ ij, & i \neq j. \end{cases}$$

Find all eigenvalues of $\mathbf{A}$ and compute $\det(\mathbf{A})$.

2011.0.2. Let D be the closed unit disk centered at the origin in the coordinate plane.

(a) Evaluate

$$\iint_D \frac{1}{\sqrt{1-x^2-y^2}}\,\mathrm{d}x\,\mathrm{d}y.$$

(b) Suppose that n rectangular pieces of paper are given, each of width 2011 and with respective heights $a_1, \ldots, a_n$. Prove that if

$$\sum_{i=1}^n a_i < 2,$$

then these pieces cannot cover D. Overlap is allowed, but folding and tearing are not.

2011.2.1. Let

$$f(x) = \frac{1}{x - x^{3/5}}.$$

Evaluate

$$\int_{2^5}^{3^5} f(x)\ \mathrm{d}x.$$

2011.2.2. For an arbitrary $n \times n$ real matrix $\mathbf{A}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{A}^2 \\ \mathbf{A}^3 & \mathbf{A}^4 \end{pmatrix} = 0.$$

2011.2.3. Let

$$f(x) = e^{\cos(x^2)}.$$

Find

$$f^{(8)}(0).$$

2011.2.4. Let (y_n) be defined by

$$y_1 = 1, \qquad y_2 = 3,$$

and, for every positive integer n,

$$y_{n+2} = \frac{2}{n}\left((n^2 + 3n + 1)y_{n+1} - 2(n+1)^2 y_n\right).$$

Prove that every term of the sequence is a positive integer.

2011.2.5. Let f be a cubic polynomial. Prove that, for every positive integer n,

$$\int_0^{2n} f(x)\ \mathrm{d}x = \frac{1}{3}(f(0) + 4f(1) + 2f(2) + 4f(3) + 2f(4) + \cdots + 2f(2n-2) + 4f(2n-1) + f(2n)).$$

2011.2.6. For a continuous function $f\colon [0,1] \to [0,\infty)$, define

$$I(f) = \int_0^1 \left(x^2 f(x) - f(x)^3\right)\ \mathrm{d}x.$$

Find the maximum possible value of $I(f)$.

2011.1.1. Define

$$f(x) = \int_{\sin x}^{1+x} e^{t^2+2xt}\ \mathrm{d}t.$$

Find $f'(0)$.

2011.1.2. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix all of whose entries are positive real numbers, and suppose that

$$\sum_{i=1}^{n} a_{ij} = 1$$

for every $j = 1, 2, \ldots, n$. Prove that every root of

$$\det(\mathbf{A} - x\mathbf{I}_n) = 0$$

has absolute value at most 1.

2011.1.3. Evaluate

$$\int_0^\infty x^3 \pi^{-\sqrt[3]{x^8}} \, \mathrm{d}x.$$

2011.1.4. For each positive integer n, let

$$A(n) = \left| \{(x, y) \in \mathbb{Z}^2 \mid x^2 + 3y^2 = n\} \right|.$$

Prove that

$$\left| \{(x, y) \in \mathbb{Z}^2 \mid x^2 + 3(4y^2 - 3y) = n\} \right| = \frac{1}{2} A(16n + 27).$$

2011.1.5. Let $f \colon \mathbb{R} \to \mathbb{R}$ be twice differentiable and satisfy $f(0) = 0$. Prove that

$$\int_0^1 |f'(x)|^2 \, \mathrm{d}x \geq \frac{1}{4} \int_0^1 \frac{|f(x)|^2}{x^2} \, \mathrm{d}x.$$

2011.1.6. Let a line ℓ and a plane P in three-dimensional Euclidean space meet at the origin. Let f be a rotation about the axis ℓ, and let g be reflection across the plane P. Prove that there exist a line ℓ' and a plane P' perpendicular to each other, together with a rotation f' about the axis ℓ' and a reflection g' across P', such that

$$f \circ g = f' \circ g'.$$

2012.2.1. Let

$$\mathbf{A} = \begin{pmatrix} -2 & 1 & 2 \\ -4 & 3 & 2 \\ -4 & 1 & 4 \end{pmatrix}.$$

Find the sum of all entries of $\mathbf{A}^{2012}$.

2012.2.2. Find all continuous functions $f \colon [0, 1] \to [0, \infty)$ satisfying

$$\int_0^1 x f(x) \, \mathrm{d}x = 3 \int_0^1 x^2 f(x) \, \mathrm{d}x = 9 \int_0^1 x^3 f(x) \, \mathrm{d}x.$$

2012.2.3. The region in the coordinate plane bounded by the line $y = x$ and the curve $y = x^2$ is rotated about the line $y = x$. Find the volume of the resulting solid.

2012.2.4. Let

$$P = (0, 1), \qquad Q = (2\sqrt{3}, 3).$$

For a point X on the x-axis, find the minimum value of

$$|X - P| + \sqrt{3}\,|X - Q|.$$

2012.2.5. Find all possible values of $\mathrm{tr}(\mathbf{A})$ for $n \times n$ matrices $\mathbf{A}$ satisfying the following conditions.

(i) $\mathbf{A}^2 = \mathbf{I}_n$.

(ii) $\mathrm{rk}(\mathbf{A} + \mathbf{I}_n) = 1$.

2012.2.6. For real numbers x, y, z, define

$$m_k = \frac{1}{3}(x^k + y^k + z^k).$$

If $m_1 = 0$, prove that

$$m_3^2 \leq m_2(m_4 - m_2^2).$$

2012.2.7. Define

$$e_n = \left(1 + \frac{1}{n}\right)^n.$$

Evaluate

$$\lim_{n\to\infty} \left(\frac{2n(e - e_n)}{e}\right)^n.$$

2012.2.8. Let $\mathbf{A}$ be a real symmetric matrix satisfying the following conditions.

(i) Every diagonal entry of $\mathbf{A}$ is 0.
(ii) Some entry of $\mathbf{A}$ is greater than 1.

Prove that $\mathbf{A}$ has an eigenvalue greater than 1.

2012.1.1. Evaluate

$$J = \int_0^1 \sqrt[3]{2x^3 - 3x^2 - x + 1}\ \mathrm{d}x.$$

2012.1.2. Prove that there do not exist square matrices $\mathbf{A}$ and $\mathbf{B}$ satisfying

$$(\mathbf{AB})^{2012} - (\mathbf{BA})^{2012} = \mathbf{I}.$$

2012.1.3. Evaluate the right-hand limit

$$\lim_{x\to 0^+} \frac{\sqrt{\tan x - x}}{\sin\sqrt{x} - \sqrt{x}}.$$

2012.1.4. For a fixed integer $n \geq 2$, find all possible values of $\mathrm{tr}(\mathbf{A})$ for $n \times n$ matrices $\mathbf{A}$ satisfying the following conditions.

(i) $\mathrm{rk}(\mathbf{A} + \mathbf{I}_n) = 1$.
(ii) $\mathrm{tr}(\mathbf{A}) = \mathrm{tr}(\mathbf{A}^3)$.

2012.1.5. Let $f\colon \mathbb{R} \to \mathbb{R}$ be differentiable and suppose that

$$2\int_0^{1/2} f(x)\ \mathrm{d}x = \int_{1/2}^1 f(x)\ \mathrm{d}x.$$

Prove that

$$3\int_0^1 \left(f'(x)\right)^2\ \mathrm{d}x \geq \left(2\int_0^1 f(x)\ \mathrm{d}x\right)^2.$$

2012.1.6. Let $\mathbf{A}$ be a square matrix with integer entries satisfying the following conditions.

(i) For an odd prime p, every entry of $\mathbf{A} - \mathbf{I}$ is divisible by p.

(ii) There exists a positive integer k such that $\mathbf{A}^k = \mathbf{I}$.

Prove that $\mathbf{A} = \mathbf{I}$.

2012.1.7. For a real sequence $X = (x_i)_{i=1}^n$, define

$$m_k(X) = \frac{1}{n}\sum_{i=1}^{n} x_i^k.$$

If $m_1(X) = 0$, prove that

$$\left(m_3(X)\right)^2 \le m_2(X)\left(m_4(X) - \left(m_2(X)\right)^2\right).$$

2012.1.8. Let $f\colon \mathbb{R} \to \mathbb{R}$ be twice differentiable, with $f(0) = 0$, and suppose that

$$\left(f'(x)\right)^2 \le f(x)f''(x) \qquad (x \in [0,1]).$$

Prove that $f(1) = 0$.

2013.0.1. Let $\mathbf{A}$ be an $n \times n$ real matrix whose row vectors all have norm 1. Prove that if

$$\operatorname{tr}(\mathbf{A}) > \sqrt{n(n-1)},$$

then

$$\det(\mathbf{A}) \neq 0.$$

2013.2.1. Evaluate the right-hand limit

$$\lim_{a\to 0^+} \sqrt{a} \iiint_{\mathbb{R}^3} \frac{1}{\left(\sqrt{x^2+y^2+z^2}+a\right)^{7/2}}\, \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z.$$

2013.2.2. Let $n \ge 2$ be an integer and let $a, b \in \mathbb{R}$. Let $\mathbf{M}_n(a,b)$ be the $n \times n$ matrix whose diagonal entries are all a and whose off-diagonal entries are all b. Find

$$\det\bigl(\mathbf{M}_n(a,b)\bigr).$$

2013.2.3. In $\mathbb{R}^3$, let

$$A = (2,0,-1), \quad B = (-1,3,-1), \quad C = (-3,1,1),$$
$$P = (-2,1,1), \quad Q = (1,1,-2).$$

(a) Prove that the line segment PQ meets the interior of triangle ABC.

(b) Find the volume of the convex hexahedron with vertices A, B, C, P, Q whose faces are all triangles.

2013.2.4. Prove that

$$\int_{\mathbb{R}} \frac{x^2}{x^4+1}\,\mathrm{d}x = \int_{\mathbb{R}} \frac{1}{x^4+1}\,\mathrm{d}x,$$

and find their common value.

2013.2.5. For arbitrary $n \times n$ matrices $\mathbf{A}, \mathbf{B}$, prove that

$$\det(\mathbf{I}_n + \mathbf{AB}) = \det(\mathbf{I}_n + \mathbf{BA}).$$

2013.2.6. Choose four distinct points at random on the unit circle and form the quadrilateral having these points as its vertices. Find the average area of such quadrilaterals.

2013.2.7. Prove that there exists a positive constant C, independent of x, such that

$$\sum_{n=1}^{\infty} \frac{|x|}{n^2(|n-x|+1)^2} < C$$

for every $x \in \mathbb{R}$.

2013.1.1. Let $\mathbf{A}$ be a 2013×2013 matrix such that each row consists of distinct positive integers not exceeding 2013. Prove that $\det(\mathbf{A})$ is divisible by 2013.

2013.1.2. Find the volume of the region

$$\left\{(x,y,z) \in \mathbb{R}^3 \mid (x^2+y^2+4z^2+3)^2 \le 16(x^2+y^2)\right\}.$$

2013.1.3. For each positive integer n, let x_n be the unique nonnegative real solution of

$$x^n + x^{n-1} + x - 1 = 0.$$

Prove that the sequence (x_n) is increasing and converges to 1.

2013.1.4. For arbitrary $n \times n$ real matrices $\mathbf{A}, \mathbf{B}$, prove that

$$\mathrm{rk}(\mathbf{I}_n + \mathbf{AB}) = \mathrm{rk}(\mathbf{I}_n + \mathbf{BA}).$$

2013.1.5. Define

$$S = \{n \in \mathbb{Z} \mid 2x^2 + xy + 4y^2 = n \text{ for some } x, y \in \mathbb{Z}\}.$$

Prove the following.

(a) If $m \in S$, then $2m \in S$.

(b) If $m, n \in S$, then $mn \in S$.

2013.1.6. Let $f\colon \mathbb{R} \to \mathbb{R}$ be twice differentiable and suppose that, for every $x \in [0,1]$,

$$f(x) \ge 0, \qquad f''(x) \le 0.$$

Prove that

$$\int_0^1 xf(x)\,\mathrm{d}x \le \frac{2}{3}\int_0^1 f(x)\,\mathrm{d}x.$$

2013.1.7. Let $f\colon [1,\infty) \to (1,\infty)$ be monotonically increasing and suppose that, for every $x \geq 1$,

$$f(x)^2 \leq f(4x) \leq 2013^{\sqrt{x}}.$$

Evaluate

$$\lim_{x\to\infty} \frac{\ln\ln f(x)}{\ln x}.$$

2014.0.1. Let $f\colon \mathbb{R} \to \mathbb{R}$ be twice differentiable and satisfy $f(0) = 0$. Prove that

$$\int_0^1 |f(x)f'(x)|\ \mathrm{d}x \leq \frac{1}{2}\int_0^1 |f'(x)|^2\ \mathrm{d}x.$$

2014.0.2. Let $f\colon [a,b] \to \mathbb{R}$ be continuous, with $a < b$, and suppose that

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for every $t \in [0,1]$ and $x, y \in [a,b]$. Prove that

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a}\int_a^b f(x)\ \mathrm{d}x \leq \frac{f(a)+f(b)}{2}.$$

2014.2.1. For a positive integer n, let

$$f(x) = \frac{e^x}{x-1},$$

and define the polynomial $P_n(x)$ by

$$f^{(n)}(x) = \frac{P_n(x)e^x}{(x-1)^{n+1}}.$$

Find $P_n(1)$ and $P_n'(1)$.

2014.2.2. Let (a_n) and (b_n) be real sequences satisfying

$$\lim_{n\to\infty}(a_n + b_n) = 0, \qquad \lim_{n\to\infty}(e^{a_n} + e^{b_n}) = 2.$$

Prove that (a_n) converges.

2014.2.3. Let m be a positive integer. Prove that every $n \times n$ real matrix $\mathbf{A}$ satisfying

$$\mathbf{A}^m = \mathbf{0}$$

also satisfies

$$\mathrm{rk}(\mathbf{A}) \leq \frac{m-1}{m}n.$$

2014.2.4. Evaluate

$$\iiint_{\mathbb{R}^3} e^{-3x^2-3y^2-2z^2-2xz+2yz}\ \mathrm{d}x\ \mathrm{d}y\ \mathrm{d}z.$$

2014.2.5. Let $n \geq 2$ be an integer, and let $M_n(\mathbb{R})$ denote the real vector space of all $n \times n$ real matrices. Let $V \subseteq M_n(\mathbb{R})$ satisfy the following conditions, where $\mathbf{E}_{ij}$ denotes the matrix whose (i,j)-entry is 1 and whose other entries are 0.

(i) V is a subspace of $M_n(\mathbb{R})$.

(ii) If $\mathbf{A}, \mathbf{B} \in V$, then $\mathbf{AB} \in V$.

(iii)
$$\mathbf{E}_{12} + \mathbf{E}_{23} + \cdots + \mathbf{E}_{n-1,n} \in V,$$
and
$$\mathbf{E}_{21} + \mathbf{E}_{32} + \cdots + \mathbf{E}_{n,n-1} \in V.$$

Prove that
$$V = M_n(\mathbb{R}).$$

2014.2.6. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real symmetric matrices all of whose eigenvalues are positive. Prove that, for every $n \times n$ real matrix $\mathbf{C}$, the block matrix
$$\begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^{\mathsf{T}} & -\mathbf{B} \end{pmatrix}$$
is invertible.

2014.1.1. Evaluate
$$\lim_{n\to\infty} \left(\frac{1}{n!}\right)^{1/(n \ln n)}.$$

2014.1.2. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real matrices, and for $t \in \mathbb{R}$ define
$$\mathbf{C}_t = \begin{pmatrix} \mathbf{A} & t^2\mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix}.$$

Prove that
$$\det(\mathbf{C}_t) = \det(\mathbf{A} + t\mathbf{B}) \det(\mathbf{A} - t\mathbf{B}).$$

2014.1.3. Find all continuous functions $f\colon [0,\infty) \to (0,\infty)$ satisfying
$$f(x)f(y) = \max\{f(t) \mid |x-y| \leq t \leq x+y\}$$
for all $x, y \geq 0$.

2014.1.4. Let $M_n(\mathbb{C})$ be the complex vector space of all $n \times n$ complex matrices. Suppose $\mathbf{A} \in M_n(\mathbb{C})$ satisfies
$$\mathbf{A}^6 - \mathbf{A}^3 + \mathbf{I}_n = \mathbf{0}.$$

Define the linear map
$$T\colon M_n(\mathbb{C}) \to M_n(\mathbb{C}), \qquad T(\mathbf{B}) = \mathbf{AB}.$$

Prove that T is diagonalizable.

2014.1.5. Let n be a positive integer, and let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real matrices satisfying
$$x\mathbf{A} + y\mathbf{B} = \mathbf{I}_n, \qquad \mathbf{AB} = \mathbf{0},$$
where $x, y \in \mathbb{R} \setminus \{0\}$. Prove that
$$\det(\mathbf{A} + \mathbf{B}) = \frac{1}{x^{\mathrm{rk}(\mathbf{A})} y^{\mathrm{rk}(\mathbf{B})}}.$$

2014.1.6. Let α be a positive irrational number and define

$$q_n = \frac{\lfloor n\alpha \rfloor}{n} \qquad (n \geq 1).$$

Prove that the sequence (q_n) is not monotonically increasing.

2015.2.1. Evaluate

$$\int_0^\infty \left(x^2 + 1 - x\sqrt{x^2+2}\right)\,\mathrm{d}x.$$

2015.2.2. For a positive integer n, let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \max\{i, j\}.$$

Find $\det(\mathbf{A})$.

2015.2.3. Determine whether the series

$$\sum_{n=1}^{\infty} \left((2n+1)e^{-1/n} - (2n-1)\right)$$

converges.

2015.2.4. Let $\mathbf{A}$ be a 5×5 matrix with rational entries having the eigenvector

$$(1, \sqrt{2}, \sqrt{3}, \sqrt{4}, \sqrt{5})^{\mathsf{T}}.$$

Prove that $\mathbf{A}^{\mathsf{T}}$ has an eigenvector all of whose entries are rational.

2015.2.5. Let $f\colon (-1, \infty) \to \mathbb{R}$ be infinitely differentiable and suppose that, for every positive integer n,

$$f\left(\frac{1}{n}\right) = \frac{n}{n+1}.$$

Find $f'''(0)$.

2015.2.6. Let $f\colon [a, b] \to \mathbb{R}$ be continuous. Prove that there exists a constant $c \in [a, b]$ such that

$$\int_a^b f(x)e^x\,\mathrm{d}x = e^a \int_a^c f(x)\,\mathrm{d}x + e^b \int_c^b f(x)\,\mathrm{d}x.$$

2015.2.7. Let n be a positive integer, and let $\mathbf{A} = (a_{ij})$ be an $n \times n$ complex matrix satisfying the following conditions.

(i) $a_{ii} = 1$ for $1 \leq i \leq n$.
(ii) $a_{ij} = 0$ for $1 \leq j < i \leq n$.
(iii) $\overline{\mathbf{A}}\mathbf{A} = \mathbf{I}_n$.

Prove that there exists an $n \times n$ complex matrix $\mathbf{B} = (b_{ij})$ satisfying the following conditions.

(i) $b_{ii} = 1$ for $1 \leq i \leq n$.
(ii) $b_{ij} = 0$ for $1 \leq j < i \leq n$.
(iii) $\overline{\mathbf{B}}\mathbf{A} = \mathbf{B}$.

Here $\overline{\mathbf{A}} = (\overline{a_{ij}})$.

2015.2.8. Let n be a positive integer, and let $\mathbf{X} = (x_{ij})$ be an $n \times n$ real matrix whose entries vary over a domain on which $\mathbf{X}$ is invertible. Write

$$\mathbf{Y} = \mathbf{X}^{-1} = (y_{pq}),$$

and regard each entry as a function

$$y_{pq} = y_{pq}(x_{11}, x_{12}, \ldots, x_{nn}).$$

Prove that

$$\frac{\partial y_{pq}}{\partial x_{ij}} = -y_{pi}y_{jq}.$$

2015.1.1. Evaluate

$$\lim_{n\to\infty} \left(1 + \frac{\ln n}{n-1}\right)^n.$$

2015.1.2. Let d be a positive integer and let

$$\phi(x, y) = \sum_{i=0}^{2d} a_i x^i y^{2d-i}$$

be an arbitrary polynomial with real coefficients. Prove that

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)^d \left(y\frac{\partial \phi}{\partial x}\right) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)^d \left(x\frac{\partial \phi}{\partial y}\right).$$

2015.1.3. Let $n \geq 3$ be an integer, and let $\mathbf{A} = (a_{ij})$ be the $n^2 \times n^2$ matrix defined by

$$a_{ij} = \begin{cases} 1, & n \mid i + j, \\ 0, & n \nmid i + j. \end{cases}$$

Find all eigenvalues of $\mathbf{A}$ and the dimension of the eigenspace corresponding to each eigenvalue.

2015.1.4. Let $V = M_{100}(\mathbb{R})$, the real vector space of all 100×100 real matrices. For $\mathbf{A} \in V$, let

$$d_{\mathbf{A}} = \dim\{\mathbf{B} \in V \mid \mathbf{AB} = \mathbf{BA}\}.$$

Suppose that $\mathbf{A} \in V$ satisfies

$$\mathbf{A}^4 - 5\mathbf{A}^2 + 4\mathbf{I} = \mathbf{0}.$$

Find the minimum possible value of $d_{\mathbf{A}}$.

2015.1.5. Let $f\colon \mathbb{R} \to \mathbb{R}$ be differentiable and satisfy

$$\lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = 0,$$

$$\int_{-\infty}^{\infty} |f(x)|^2 \, \mathrm{d}x < \infty, \qquad \int_{-\infty}^{\infty} |f'(x)|^2 \, \mathrm{d}x < \infty.$$

Prove that

$$\max_{x\in\mathbb{R}} |f(x)| \leq \left(\int_{-\infty}^{\infty} |f(x)|^2 \, \mathrm{d}x\right)^{1/4} \left(\int_{-\infty}^{\infty} |f'(x)|^2 \, \mathrm{d}x\right)^{1/4}.$$

2015.1.6. Let $x(t), y(t), z(t)$ be relatively prime polynomials with real coefficients, each of degree at least 1. Suppose that, for some positive integer d,

$$x(t)^d + y(t)^d = z(t)^d.$$

Prove that $x(t)^{d-1}$ divides

$$y(t)z'(t) - z(t)y'(t),$$

and that $d \leq 2$.

2015.1.7. Let $\mathbf{A}$ and $\mathbf{B}$ be 2×2 real symmetric matrices with eigenvalues

$$\lambda_1 \geq \lambda_2 \geq 0 \qquad \text{and} \qquad \mu_1 \geq \mu_2 \geq 0,$$

respectively. Prove that

$$\operatorname{tr}(\mathbf{AB}) \leq \lambda_1\mu_1 + \lambda_2\mu_2.$$

2015.1.8. For a positive integer n, call an $n \times n$ real matrix $\mathbf{M}$ orthogonal if

$$\mathbf{M}^\mathsf{T}\mathbf{M} = \mathbf{I}_n,$$

and call $\mathbf{M}$ *similar to an orthogonal matrix* if there exists an invertible real matrix $\mathbf{P}$ such that

$$\mathbf{PMP}^{-1}$$

is orthogonal.

(a) Let $\mathbf{S}$ be a real symmetric matrix all of whose eigenvalues are positive. Prove that every matrix $\mathbf{A}$ satisfying

$$\mathbf{A}^\mathsf{T}\mathbf{SA} = \mathbf{S}$$

is similar to an orthogonal matrix.

(b) Let $\mathbf{A}$ and $\mathbf{B}$ be matrices such that both $\mathbf{A}$ and

$$\begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{pmatrix}$$

are similar to orthogonal matrices. Prove that $\mathbf{B}$ is also similar to an orthogonal matrix.

2016.0.1. Using ln for the natural logarithm, evaluate

$$\lim_{n\to\infty} \frac{1}{n} \ln\left(\sum_{k=2}^{2^n} k^{1/n^2}\right).$$

2016.0.2. Let $f\colon [-\pi/4, \pi/4] \to [-1, 1]$ be continuous and differentiable on $(-\pi/4, \pi/4)$. Prove that there exists $x_0 \in (-\pi/4, \pi/4)$ such that

$$|f'(x_0)| \leq 1 + f(x_0)^2.$$

2016.0.3. Let m, n be positive integers, and let $\mathbf{A}$ be an $m \times n$ real matrix. Define

$$\mathbf{X} = \mathbf{I}_m + \mathbf{AA}^\mathsf{T}, \qquad \mathbf{Y} = \mathbf{I}_n + \mathbf{A}^\mathsf{T}\mathbf{A}.$$

(a) Prove that $\mathbf{X}$ and $\mathbf{Y}$ are invertible.

(b) Find

$$\operatorname{tr}(\mathbf{X}^{-1}) - \operatorname{tr}(\mathbf{Y}^{-1}).$$

2016.2.1. Let $a, b \in \mathbb{R}$ satisfy

$$a^2 + b^2 \leq \frac{\pi^2}{4}.$$

Prove that

$$\cos\sqrt{a^2+b^2} \leq \cos a \cos b.$$

2016.2.2. Let $\mathbf{A}$ be an $n \times n$ real matrix having n distinct positive eigenvalues. Find the number of real matrices $\mathbf{B}$ satisfying

$$\mathbf{B}^{2016} = \mathbf{A}.$$

2016.2.3. Let $y(t)$ satisfy

$$y'(t) = y(t) - \int_0^t (y'(s))^2 \cos s \, \mathrm{d}s, \qquad y(0) = 2.$$

Find $y(\pi/4)$.

2016.2.4. Let $x_1, x_2, x_3 > 0$ and let $a \in [-\frac{1}{2}, 1]$. Define

$$m = \frac{x_1 + x_2 + x_3}{3},$$

$$y_1 = (1-a)m + ax_1, \qquad y_2 = (1-a)m + ax_2, \qquad y_3 = (1-a)m + ax_3.$$

Prove that

$$\frac{27y_1y_2y_3}{(y_1+y_2+y_3)^3} \geq \left(\frac{27x_1x_2x_3}{(x_1+x_2+x_3)^3}\right)^{\max\{a,-2a\}}.$$

2016.2.5. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real matrices satisfying

$$\mathbf{A} \neq \mathbf{0}, \qquad \mathbf{AB} - \mathbf{BA} = \mathbf{A}.$$

Prove that

$$\mathbf{A}^n = \mathbf{0} \qquad \text{and} \qquad \mathbf{B}^n \neq \mathbf{0}.$$

2016.1.1. Let $(a_n)_{n\geq 1}$ be a decreasing real sequence satisfying

$$\lim_{n\to\infty} a_n = 0.$$

For

$$S_m = \sum_{n=1}^{m} a_n,$$

suppose that

$$S_{2^k} - 2^k a_{2^k} \leq 1$$

for every positive integer k. Prove that

$$\sum_{n=1}^{\infty} a_n \leq 1.$$

2016.1.2. Let

$$f(x) = \cos\left(\frac{3\sqrt{3}\,\pi}{8}(x - x^3)\right).$$

Evaluate

$$\lim_{t\to\infty}\left(\int_0^1 f(x)^t\,\mathrm{d}x\right)^{1/t} + \lim_{t\to-\infty}\left(\int_0^1 f(x)^t\,\mathrm{d}x\right)^{1/t}.$$

2016.1.3. Let $\mathbf{A}$ and $\mathbf{B}$ be 2×2 real matrices satisfying

$$\det(\mathbf{A}) = \det(\mathbf{B}) = 1, \qquad \mathrm{tr}(\mathbf{A}) > 2, \qquad \mathrm{tr}(\mathbf{B}) > 2,$$

and

$$\mathrm{tr}\Big(\mathbf{A}\mathbf{B}\mathbf{A}^{-1}\mathbf{B}^{-1}\Big) = 2.$$

Prove that $\mathbf{A}$ and $\mathbf{B}$ have a common eigenvector.

2016.1.4. Let $M > 0$, and let $f\colon [0,\infty) \to [0, M]$ be continuous and satisfy

$$\int_0^\infty (1+x)f(x)\,\mathrm{d}x < \infty.$$

Prove that

$$\left(\int_0^\infty f(x)\,\mathrm{d}x\right)^2 \le 4M\int_0^\infty xf(x)\,\mathrm{d}x.$$

2016.1.5. Let $\mathbf{A}$ be an $n\times n$ complex matrix. Prove that the following conditions are equivalent.

(i) There exists an $n\times n$ complex matrix $\mathbf{B}$ such that

$$\mathbf{A}\mathbf{B} - \mathbf{B}\mathbf{A} = \mathbf{A}.$$

(ii) There exists a positive integer k such that

$$\mathbf{A}^k = \mathbf{0}.$$

2017.0.1. Let (a_n) be a decreasing sequence of positive numbers, and let

$$0 < \theta < \frac{\pi}{2}.$$

Prove that

$$\left|\sum_{n=1}^{2017} a_n\cos(n\theta)\right| \le \frac{\pi a_1}{\theta}.$$

2017.2.1. Let V be the real vector space of all real polynomials of degree at most 3, and define a linear map $T\colon V\to V$ by letting $T(P(x))$ be the remainder when $x^4P(x)$ is divided by

$$(x-1)^2(x+1)^2.$$

Find the characteristic polynomial of T.

2017.2.2. Find the solution $u(t)$ of

$$\begin{cases} u'(t) = -u(t) + u(t)^2 e^t, & t > 0, \\ u(0) = -1. \end{cases}$$

2017.2.3. Let n be a positive integer, and let $\mathbf{A}$ be an $n \times n$ real matrix satisfying

$$\mathbf{A}^{n+1} = \mathbf{0}.$$

Prove that

$$\mathbf{A}^n = \mathbf{0}.$$

2017.2.4. Let $f(x, y)$ be a polynomial with rational coefficients such that $f(m, n)$ is an integer for every pair of integers m, n. Prove that there exist integers c_{ij} such that

$$f(x, y) = \sum_{i,j \geq 0} c_{ij} \binom{x}{i} \binom{y}{j},$$

where

$$\binom{x}{i} = \frac{x(x-1)\cdots(x-i+1)}{i!}, \qquad \binom{x}{0} = 1.$$

2017.2.5. Let

$$\mathbf{A} = \begin{pmatrix} 2 & 2 & 3 \\ 0 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix}.$$

Prove that if a real matrix $\mathbf{B}$ satisfies

$$\mathbf{AB} = \mathbf{BA},$$

then there exist real numbers a, b, c such that

$$\mathbf{B} = a\mathbf{A}^2 + b\mathbf{A} + c\mathbf{I}.$$

2017.2.6. Let $n \geq 3$ be a positive integer, and let $\mathbf{A} = (a_{ij})$ be an $n \times n$ real matrix satisfying the following conditions.

(i) $a_{ij} > 0$ for all i, j.

(ii) For every i,

$$2a_{ii} = \sum_{j=1}^{n} a_{ij}.$$

Prove that $\mathbf{A}$ is invertible.

2017.2.7. Evaluate

$$\lim_{n \to \infty} n \int_0^{\pi/2} (1 - \sin x)^n \, \mathrm{d}x.$$

2017.1.1. Let $S = \{1, 2, \ldots, 2017\}$, and let T be the set of all $n \times n$ matrices whose entries belong to S, where $n \geq 2$ is a positive integer. Evaluate

$$\sum_{\mathbf{A} \in T} \det(\mathbf{A}).$$

2017.1.2. Prove that every polynomial $f(x, y)$ with real coefficients can be expressed as a real linear combination of polynomials of the form

$$(x + ay)^k,$$

where a is an arbitrary real number and k is a nonnegative integer.

2017.1.3. Let (a_n) be a sequence satisfying $a_1 > 1$ and

$$a_{n+1} = 1 + \frac{n^2}{a_n}.$$

Evaluate

$$\lim_{n \to \infty} \frac{a_n}{n}.$$

2017.1.4. Let

$$f(x) = x^3 + ax^2 + bx + c$$

be a cubic polynomial with real coefficients, and let α, β, γ be the three roots of $f(x) = 0$. Prove that α, β, γ are three distinct real numbers if and only if the real symmetric matrix

$$\begin{pmatrix} 3 & p_1 & p_2 \\ p_1 & p_2 & p_3 \\ p_2 & p_3 & p_4 \end{pmatrix}$$

is positive definite, where

$$p_i = \alpha^i + \beta^i + \gamma^i.$$

2017.1.5. Evaluate

$$\lim_{n \to \infty} \sqrt{n} \int_0^\pi \sin^n x \, \mathrm{d}x.$$

2017.1.6. Let n be a positive integer, let $\mathbf{A}$ be an $n \times n$ real matrix, and let $\mathbf{J}$ be the $n \times n$ all-ones matrix. Suppose that

$$\mathbf{A} + \mathbf{A}^\mathsf{T} = \frac{1}{n}\mathbf{J}, \qquad \mathbf{AJ} = \frac{1}{2}\mathbf{J}.$$

Prove that $\mathbf{A}^m - \mathbf{I}$ is invertible for every positive odd integer m.

2017.1.7. Let $u(t)$ be the solution of

$$\begin{cases} u''(t) + u'(t) = \sin u(t), & t > 0, \\ u(0) = 1, \quad u'(0) = 0. \end{cases}$$

(a) Prove that $u(t)$ and $u'(t)$ are bounded for $t > 0$.

(b) Evaluate

$$\lim_{t \to \infty} u(t).$$

2018.0.1. Let $(\alpha_1, \ldots, \alpha_n)$ and $(\beta_1, \ldots, \beta_n)$ be arbitrary permutations of $1, 2, \ldots, n$, where $n \geq 2$. Define the $n \times n$ matrix $\mathbf{A} = (a_{ij})$ by

$$a_{ij} = (1 + \alpha_i \beta_j)^{n-1}.$$

Find all possible values of $\det(\mathbf{A})$.

2018.2.1. Let $\theta(h) \in [0, \pi/2]$ be the angle between the two lines

$$y = (1 - h)x \qquad \text{and} \qquad y = (1 + h)x.$$

Evaluate

$$\lim_{h \to 0^+} \frac{\theta(h)}{h}.$$

2018.2.2. Find the solution of

$$\begin{cases} \dfrac{\mathrm{d}y}{\mathrm{d}x} = x^3(x + y)^2 - \dfrac{3(x + y)}{x} - 1, \\ y(2) = -3. \end{cases}$$

2018.2.3. Let V be the real vector space of all 2×2 real matrices. Let $\mathbf{A} \in V$ satisfy

$$\operatorname{tr}(\mathbf{A}) = 3, \qquad \operatorname{tr}(\mathbf{A}^2) = 5.$$

Define a linear map $T \colon V \to V$ by

$$T(\mathbf{B}) = \mathbf{AB} + \mathbf{BA}.$$

Find

$$\operatorname{tr}(T^3).$$

2018.2.4. Two points A and B on the Northern Hemisphere of the Earth have positions (latitude α, longitude 0) and (latitude β, longitude $\pi/2$), respectively. Assume that the Earth is a sphere, angles are measured in radians, and

$$0 < \alpha, \beta < \frac{\pi}{2}.$$

Travel from A to B along the great circle through the two points while remaining on the Northern Hemisphere. Let γ be the longitude at the point of maximum latitude along this path. Express $\tan \gamma$ in terms of α and β.

2018.2.5. Evaluate

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^{n+k-1}}{n^{k+2} 2^{m(k+1)}}.$$

2018.2.6. For each positive integer n, define

$$a_n = \frac{1}{(1 + \sqrt[3]{2})^n} \sum_{k=0}^{\infty} \binom{n}{3k+2} 2^k.$$

Evaluate

$$\lim_{n \to \infty} a_n.$$

Here $\binom{n}{3k+2} = 0$ when $3k + 2 > n$.

2018.2.7. Let $f\colon \mathbb{R} \to (0,\infty)$ be twice differentiable and satisfy

$$\int_{-\infty}^{\infty} f(x)\ \mathrm{d}x = 1, \qquad \int_{-\infty}^{\infty} xf(x)\ \mathrm{d}x = 0, \qquad \int_{-\infty}^{\infty} x^2 f(x)\ \mathrm{d}x = 1.$$

Prove that

$$\int_{-\infty}^{\infty} \frac{f'(x)^2}{f(x)}\ \mathrm{d}x \geq 1,$$

and find all functions f for which equality holds.

2018.1.1. Let

$$\mathbf{u} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \in \mathbb{R}^3, \qquad \mathbf{v}_0 = (1,2,3),$$

and define

$$\mathbf{v}_n = \mathbf{u} \times \mathbf{v}_{n-1}$$

for every positive integer n. Evaluate

$$\sum_{n=1}^{\infty} (3,2,1) \cdot \mathbf{v}_{2n}.$$

2018.1.2. Let n be a positive integer, and let $\mathbf{A}$ be an $n \times n$ real matrix satisfying

$$\mathrm{tr}(\mathbf{A}) = 2018.$$

Prove that if

$$\mathrm{rk}(\mathbf{A}) = 1,$$

then

$$\mathbf{A}^2 = 2018\mathbf{A}.$$

2018.1.3. For a region S in the coordinate plane $\mathbb{R}^2$, suppose that the following integral is well-defined, and define

$$\Phi(S) = \iint_S (1 - 5x^2 + 4xy - 5y^2)\ \mathrm{d}x\ \mathrm{d}y.$$

Find the maximum possible value of $\Phi(S)$.

2018.1.4. Let $\mathbf{A} = (a_{ij})$ be a 2018×2018 real symmetric matrix satisfying

$$|a_{ij} - 2018| \leq 1$$

for every i, j. Let $\lambda(\mathbf{A})$ denote the largest eigenvalue of $\mathbf{A}$. Find the maximum and minimum possible values of $\lambda(\mathbf{A})$.

2018.1.5. Let $f\colon [0,1] \to \mathbb{R}$ be continuous, differentiable on $(0,1)$, and satisfy

$$f(0) = 1, \qquad f(1) = 0.$$

Prove that there exists a positive real number $x_0 < 1$ such that

$$|f'(x_0)| \geq 2018\, f(x_0)^{2018}.$$

2018.1.6. Let $\mathbf{A}$ be a 3×3 real matrix satisfying

$$\mathbf{v}^{\mathsf{T}}\mathbf{A}\mathbf{v} > 0$$

for every $\mathbf{v} \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$. The matrix $\mathbf{A}$ is not assumed to be symmetric.

(a) Prove that

$$\det(\mathbf{A}) > 0.$$

(b) Let

$$\mathbf{D} = \operatorname{diag}(-1, 1, 1).$$

Prove that $\mathbf{AD}$ has exactly one negative eigenvalue.

2018.1.7. Let $(a_n)_{n\geq 0}$ be a real sequence satisfying

$$a_0 = 0, \qquad \frac{100}{101} < a_{100} < 1,$$

and, for every $n \geq 1$,

$$2a_n - a_{n-1} - a_{n+1} \leq 2(1 - a_n)^3.$$

(a) Define

$$b_n = a_n - \frac{n}{n+1}.$$

Prove that

$$b_n \leq b_{n+1}$$

for every $n \geq 100$.

(b) Determine whether the series

$$\sum_{n=1}^{\infty} \frac{a_n}{n^2}$$

converges or diverges.

2019.0.1. Let

$$\mathbf{F}(x, y, z) = (\sin(xy), \sin(yz), 0), \qquad \mathbf{G}(x, y, z) = (e^{x^2+y^2+z^2}, \cos(xz), 0)$$

be vector fields defined on

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 1,\ |z| \leq 1\}.$$

Evaluate

$$\iiint_W (\mathbf{G} \cdot \operatorname{curl}(\mathbf{F}) - \mathbf{F} \cdot \operatorname{curl}(\mathbf{G}))\ \mathrm{d}V.$$

2019.0.2. For a real number a and an integer $n \geq 2$, define

$$S_n(a) = n^a \sum_{k=1}^{n-1} \frac{1}{k^{2019}(n-k)^{2019}}.$$

Find all real numbers a such that the sequence $(S_n(a))_{n\geq 2}$ converges to a positive real number.

2019.2.1. Evaluate

$$\lim_{n\to\infty} n \sin\left(\sin\left(\sin\frac{2019}{n}\right)\right).$$

2019.2.2. Find the solution of

$$\begin{cases} x'(t) = x(t)(x(t)y(t)-1), \\ y'(t) = y(t)(x(t)y(t)-1), \\ x(0) = \dfrac{1}{2}, \quad y(0) = 1. \end{cases}$$

2019.2.3. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix defined by

$$a_{ij} = x^{|i-j|},$$

where x is a positive real number.

(a) Find $\det(\mathbf{A})$.

(b) Find all x for which $\mathbf{A}$ is invertible, and in that case find $\mathbf{A}^{-1}$.

2019.2.4. Let $M_n(\mathbb{R})$ be the real vector space of $n \times n$ real matrices. For $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{R})$, define a linear map

$$L_{\mathbf{A},\mathbf{B}} \colon M_n(\mathbb{R}) \to M_n(\mathbb{R})$$

by

$$L_{\mathbf{A},\mathbf{B}}(\mathbf{X}) = \mathbf{AXB}.$$

Prove that

$$\det(L_{\mathbf{A},\mathbf{B}}) = \det(\mathbf{A}^n\mathbf{B}^n).$$

2019.2.5. Let $f \colon \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \frac{\sqrt{3}(2x+1)}{(x^2+x+3)^2+3}.$$

Evaluate

$$\sum_{n=1}^{\infty}\left(\frac{\pi}{6} - \int_0^n f(x)\,\mathrm{d}x\right).$$

2019.2.6. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^4$ be mutually orthogonal vectors of lengths $2, 3, 4, 5$, respectively. Prove that for every two-dimensional subspace $W \subset \mathbb{R}^4$, at least one of the vectors obtained by orthogonally projecting $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ onto W is not a unit vector.

2019.1.1. Let

$$\mathbf{A} = \begin{pmatrix} 2019 & 2020 & 2021 \\ 2020 & 2021 & 2022 \\ 2021 & 2022 & 2023 \end{pmatrix}.$$

Find $\mathrm{rk}(\mathbf{A})$.

2019.1.2. Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 12 \end{pmatrix}.$$

Find the volume of

$$V = \{\mathbf{x} \in \mathbb{R}^3 \mid \mathbf{x} \cdot (\mathbf{Ax}) \le 1 < \mathbf{x} \cdot (\mathbf{Bx})\}.$$

2019.1.3. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ unitary matrices. Prove that

$$|\det(\mathbf{A} + 2\mathbf{B})| \leq 3^n.$$

2019.1.4. Let $(a_n)_{n\geq 1}$ be the real sequence defined by

$$a_1 = 1, \qquad a_{n+1} = \ln\left(\frac{e^{a_n} - 1}{a_n}\right),$$

and define

$$b_n = \prod_{i=1}^{n} a_i.$$

Evaluate

$$\sum_{n=1}^{\infty} b_n.$$

2019.1.5. Let $f\colon [0,\infty) \to [0,\infty)$ be integrable and satisfy

$$\int_0^\infty f(x)^2 \, \mathrm{d}x < \infty, \qquad \int_0^\infty xf(x) \, \mathrm{d}x < \infty.$$

Prove that

$$\left(\int_0^\infty f(x) \, \mathrm{d}x\right)^3 \leq 8\left(\int_0^\infty f(x)^2 \, \mathrm{d}x\right)\left(\int_0^\infty xf(x) \, \mathrm{d}x\right).$$

2019.1.6. Let $M_n(\mathbb{C})$ be the complex vector space of $n \times n$ complex matrices, and let

$$T\colon M_n(\mathbb{C}) \to M_n(\mathbb{C})$$

be a linear map satisfying

$$\det(\mathbf{A}) = \det(T(\mathbf{A}))$$

for every $\mathbf{A} \in M_n(\mathbb{C})$.

(a) Prove that if $T(\mathbf{A}) = \mathbf{0}$, then $\mathbf{A} = \mathbf{0}$.

(b) Prove that

$$\mathrm{rk}(\mathbf{A}) = \mathrm{rk}(T(\mathbf{A}))$$

for every $\mathbf{A} \in M_n(\mathbb{C})$.

2020.0.1. Evaluate

$$\lim_{x\to 0} \frac{e^x \sin(x^2) - x^2}{x^3}.$$

2020.0.2. Evaluate

$$\lim_{n\to\infty} \int_0^{\pi/2} \frac{\cos^2 x}{1 + \cos^2(nx)} \, \mathrm{d}x.$$

2020.2.1. Let $n \geq 2$ be a positive integer, and define the $n \times n$ symmetric matrix $\mathbf{M} = (m_{ij})$ by

$$m_{ij} = \begin{cases} 1, & i = 1 \text{ or } j = 1, \\ \dfrac{j-1}{j}, & i = j \geq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Express

$$n \det(\mathbf{M}) + 1 + \frac{1}{2} + \cdots + \frac{1}{n-1}$$

as a polynomial in n.

2020.2.2. Define

$$f(x, y) = 3 + 2\cos x + 2\cos y + 2\cos(x - y).$$

(a) Prove that

$$f(x, y) \geq 0$$

for all real x, y.

(b) Find all (x, y) in

$$\{(x, y) \in \mathbb{R}^2 \mid -\pi \leq x, y \leq \pi\}$$

such that

$$f(x, y) = 0.$$

2020.2.3. Determine the necessary and sufficient conditions on a, b for the three matrices

$$\mathbf{X} = \begin{pmatrix} a & 1 \\ 0 & b \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} b & -1 \\ 0 & a \end{pmatrix}, \qquad \mathbf{Z} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

to have a common eigenvector.

2020.2.4. For three vectors

$$\mathbf{w}_i = (a_{i1}, a_{i2}, a_{i3}) \in \mathbb{R}^3, \qquad i = 1, 2, 3,$$

write $|\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3|$ for the determinant of the matrix $(a_{ij})_{1 \leq i,j \leq 3}$. Prove that for arbitrary vectors

$$\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3,$$

$$\begin{aligned} |\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3||\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3| = {} & |\mathbf{v}_1, \mathbf{u}_2, \mathbf{u}_3||\mathbf{u}_1, \mathbf{v}_2, \mathbf{v}_3| \\ & + |\mathbf{u}_1, \mathbf{v}_1, \mathbf{u}_3||\mathbf{u}_2, \mathbf{v}_2, \mathbf{v}_3| \\ & + |\mathbf{u}_1, \mathbf{u}_2, \mathbf{v}_1||\mathbf{u}_3, \mathbf{v}_2, \mathbf{v}_3|. \end{aligned}$$

2020.2.5. Evaluate

$$\lim_{n \to \infty} \sum_{j=1}^{n} \sum_{k=j^2}^{n^2} \frac{j^3 \sqrt{k^3 + n^6}}{n^9}.$$

2020.2.6. Let $x(t)$ and $y(t)$ be differentiable real-valued functions on $[0,\infty)$ satisfying

$$(x(0),y(0)) = (2020,2020)$$

and, for every $t>0$,

$$\begin{cases} x'(t) = -2x(t)y(t) + 3x(t) + 1, \\ y'(t) = x(t)^2 - x(t) - y(t) + 2. \end{cases}$$

Prove that the following limit exists and find its value:

$$\lim_{t\to\infty} \big(x(t)+y(t)\big).$$

2020.1.1. Let n be a positive integer that is a perfect square and whose last four digits are all the same. Prove that n is divisible by 10000.

2020.1.2. Let $\mathbf{Z}$ be a nonzero $n\times n$ real symmetric matrix satisfying

$$\operatorname{tr}(\mathbf{Z}) \le 0.$$

Prove that for every sufficiently large positive integer k,

$$\det\left(\mathbf{I}_n + \frac{1}{k}\mathbf{Z}\right) < 1.$$

2020.1.3. Let $\mathbf{A}$ be a 2020×2020 matrix of rank 2 whose entries are all 0 or 1. Define $n(\mathbf{A})$ to be the sum of all entries of $\mathbf{A}$. Find all possible values of $n(\mathbf{A})$.

2020.1.4. Let

$$D = \left\{(x,y)\in\mathbb{R}^2 \mid |x-y| \le \frac{2}{3}, \quad 0\le x,y\le 1\right\}.$$

Evaluate

$$\iint_D (1+(x-y)^3)\,\mathrm{d}A.$$

2020.1.5. Let $g\colon \mathbb{R}\to(0,\infty)$ be twice differentiable. Suppose that $g''(x)$ is continuous at every point and

$$g'(0) = g'(1) = 0.$$

Prove that

$$\int_0^1 \frac{g'(x)^4}{g(x)^2}\,\mathrm{d}x \le 9\int_0^1 g''(x)^2\,\mathrm{d}x.$$

2020.1.6. Let $(a_{m,n})_{m\ge 0,\, n\ge 1}$ be an array of positive real numbers satisfying, for every pair of positive integers m,n,

$$a_{m+1,n}a_{m-1,n+1} = a_{m,n}a_{m,n+1} - 1, \qquad a_{0,n} = 1.$$

For each positive integer m, define the matrix $\mathbf{B}_m = (b_{ij})_{1\le i,j\le m}$ by

$$b_{ij} = \begin{cases} a_{1,i}, & i = j, \\ 1, & |i-j| = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Prove that for every positive integer m,

$$a_{m,1} = \det(\mathbf{B}_m).$$

2021.0.1. Let $f\colon \mathbb{R} \to \mathbb{R}$ be twice differentiable and satisfy

$$\lim_{x\to\infty} f(x) = 0.$$

Prove that

$$\int_0^\infty x^4 f(x)^4 \, \mathrm{d}x \le 2\left(\int_0^\infty x^2 f(x)^2 \, \mathrm{d}x\right)^{3/2} \left(\int_0^\infty x^2 f'(x)^2 \, \mathrm{d}x\right)^{1/2}.$$

2021.2.1. Evaluate

$$\lim_{x\to 0} \frac{e^{x^2} - e^{-x^2}}{1 - \cos x}.$$

2021.2.2. Solve the differential equation

$$2yy'' + 2(y')^2 - 8yy' + 3y^2 = 0, \qquad y(0) = 2, \qquad y'(0) = 3.$$

2021.2.3. Determine all integers $n \ge 2$ and all real numbers $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n$ for which the matrix

$$\begin{pmatrix} \cos(\alpha_1 - \beta_1) & \cos(\alpha_1 - \beta_2) & \cdots & \cos(\alpha_1 - \beta_n) \\ \cos(\alpha_2 - \beta_1) & \cos(\alpha_2 - \beta_2) & \cdots & \cos(\alpha_2 - \beta_n) \\ \vdots & \vdots & \ddots & \vdots \\ \cos(\alpha_n - \beta_1) & \cos(\alpha_n - \beta_2) & \cdots & \cos(\alpha_n - \beta_n) \end{pmatrix}$$

is invertible.

2021.2.4. Let

$$D = \{(x, y) \in \mathbb{R}^2 \mid x > 0,\ y > 0,\ 1 - x - y > 0\},$$

and define $f\colon D \to \mathbb{R}$ by

$$f(x, y) = 4\big(x^2 + y^2 + (1 - x - y)^2\big) - x\ln x - y\ln y - (1 - x - y)\ln(1 - x - y).$$

Find the number of critical points of f.

2021.2.5. Evaluate

$$\int_0^1 \int_0^1 \int_0^1 \lfloor x + y + z \rfloor \, \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z.$$

2021.2.6. Let $\mathbf{A} = (a_{ij})$ be an invertible $n \times n$ matrix satisfying, for every integer k with $1 \le k \le n$,

$$\sum_{i=1}^n a_{ik} = \sum_{j=1}^n a_{kj} = 2021.$$

If $\mathbf{B} = (b_{ij}) = \mathbf{A}^{-1}$, prove that, for every integer k with $1 \le k \le n$,

$$\sum_{i=1}^n b_{ik} = \sum_{j=1}^n b_{kj} = \frac{1}{2021}.$$

2021.2.7. For real numbers a, b, let

$$\mathbf{A} = \begin{pmatrix} a & 1 & 1 & 1 \\ 1 & b & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & b \end{pmatrix}.$$

Prove that $\mathbf{A}$ is positive definite if and only if $a > 1$ and $b > 1$.

2021.1.1. Let

$$f(x, y, z) = e^{x^2 y} \sin z.$$

Find

$$\nabla f\left(\sqrt{2}, \frac{1}{2}, \frac{\pi}{4}\right).$$

2021.1.2. Let $M_n(\mathbb{R})$ be the real vector space of all $n \times n$ real matrices. For $\mathbf{A} \in M_n(\mathbb{R})$, define the linear map

$$\Phi_{\mathbf{A}} \colon M_n(\mathbb{R}) \to M_n(\mathbb{R}), \qquad \Phi_{\mathbf{A}}(\mathbf{X}) = \mathbf{AX} - \mathbf{XA}.$$

Find $\det(\Phi_{\mathbf{A}})$ and $\operatorname{tr}(\Phi_{\mathbf{A}})$.

2021.1.3. Evaluate

$$\int_0^1 \int_z^1 \int_{\sqrt{y}}^1 \cos(x^5)\, \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z.$$

2021.1.4. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ real symmetric matrix, where $n \geq 2$. Suppose that

$$a_{ij} \leq 0 \qquad \text{for all } i \neq j,$$

and that every eigenvalue of $\mathbf{A}$ is nonnegative.

(a) Prove that

$$\ker \mathbf{A} = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{v} \cdot (\mathbf{Av}) = 0\}.$$

(b) Suppose that for every proper partition I, J of $\{1, 2, \ldots, n\}$, there exist $i \in I$ and $j \in J$ such that $a_{ij} \neq 0$. Prove that

$$\dim \ker \mathbf{A} \leq 1.$$

Here a proper partition I, J means that I and J are nonempty,

$$I \cup J = \{1, 2, \ldots, n\}, \qquad I \cap J = \varnothing.$$

2021.1.5. Evaluate

$$\int_0^1 \int_0^1 \lfloor x^2 + y \rfloor\, \mathrm{d}x\, \mathrm{d}y.$$

2021.1.6. Let

$$S = \{\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_k\}$$

be a set of k distinct invertible $n \times n$ matrices, where $n, k \geq 2$. Suppose that for every $\mathbf{A}_i \in S$,

$$S = \{\mathbf{A}_1\mathbf{A}_i, \mathbf{A}_2\mathbf{A}_i, \ldots, \mathbf{A}_k\mathbf{A}_i\}.$$

Prove that there exists $\mathbf{A}_t \in S$ such that

$$(\mathbf{A}_1 + \mathbf{A}_2 + \cdots + \mathbf{A}_k) - \mathbf{A}_t$$

is invertible.

2021.1.7. Let $\mathbf{A}$ and $\mathbf{B}$ be 3×3 orthogonal matrices. Suppose that each of $\mathbf{A}$ and $\mathbf{B}$ has two eigenvectors with eigenvalue -1 and at least three eigenvectors with eigenvalue 1 in the finite set

$$\{(x, y, z) \in \mathbb{R}^3 \mid x, y, z \in \mathbb{Z},\ x^2 + y^2 + z^2 \leq 2\}.$$

Prove that at least one of

$$(\mathbf{AB})^2, \qquad (\mathbf{AB})^3, \qquad (\mathbf{AB})^4$$

equals $\mathbf{I}_3$.

2023.0.1. Let $(a_n)_{n \geq 0}$ be defined by

$$\sqrt[4]{\frac{1+4x}{1-4x}} = \sum_{n=0}^{\infty} a_n x^n, \qquad |x| < \frac{1}{4}.$$

Prove that a_n is a nonnegative integer for every $n \geq 0$.

2023.2.1. Let D be the region on the unit sphere $x^2 + y^2 + z^2 = 1$ defined by

$$D = \left\{ (x, y, z) \mid x^2 + y^2 + z^2 = 1, \quad x, y \geq 0, \quad \frac{1}{2} \leq z \leq \frac{\sqrt{3}}{2} \right\}.$$

A light source is placed at $(0, 0, 1)$. The region D is projected from this light source onto the xy-plane. Find the ratio of the area of the projected region to the area of D.

2023.2.2. Find the function $y = y(x)$ defined on the open interval $(1/e, e)$ that satisfies

$$xy' = \sqrt{x^2 - y^2} + y, \qquad -x < y < x, \qquad y(1) = \frac{1}{2}.$$

2023.2.3. Let

$$f(x) = e^{x^8} \sec(x^4).$$

Find

$$\frac{f^{(16)}(0)}{16!}.$$

2023.2.4. Let $\mathbf{A} = (a_{ij})$ be a $2 \times n$ real matrix, where $n \geq 4$. For an ordered pair (p, q) with $1 \leq p, q \leq n$ and $p \neq q$, define

$$f(p, q) = \det \begin{pmatrix} a_{1p} & a_{1q} \\ a_{2p} & a_{2q} \end{pmatrix}.$$

(a) Prove that for distinct p, q, r, s,

$$f(p, q) f(r, s) = f(r, q) f(p, s) + f(s, q) f(r, p).$$

(b) Suppose that for every q with $2 \leq q \leq n$,

$$f(1, q) > 0 \qquad \text{and} \qquad f(q-1, q) > 0.$$

Prove that

$$f(p, q) > 0$$

for every ordered pair (p, q) with $1 \leq p < q \leq n$.

2023.2.5. Let $\mathbf{A}$ be an $n \times n$ real matrix. Prove that there exists a diagonal matrix

$$\mathbf{D} = \operatorname{diag}(d_1, \dots, d_n), \qquad d_1, \dots, d_n \in \{-1, 1\},$$

such that

$$\det(\mathbf{A} + \mathbf{D}) \neq 0.$$

2023.2.6. Players A and B play a game in which the first player to climb a staircase of 12 steps wins. They alternately roll a fair 12-sided die whose faces are labeled $1, 2, \dots, 12$, according to the following rules:

(1) If A rolls a multiple of 4, A climbs 4 steps.
(2) If B rolls a multiple of 6, B climbs 6 steps.
(3) Otherwise, the player does not move.

A rolls first. Find the probability that A wins. The players continue alternating rolls until one of them wins.

2023.1.1. Find the area of the region satisfying

$$2 - 2\cos\theta \leq r \leq 1$$

in polar coordinates.

2023.1.2. Let $V = M_2(\mathbb{R})$ be the vector space of all 2×2 real matrices. Define the linear map

$$T \colon V \to V, \qquad T(\mathbf{A}) = \mathbf{A}^{\mathsf{T}}.$$

Find all eigenvalues of T and all eigenvectors corresponding to each eigenvalue.

2023.1.3. A point X moves in the coordinate plane with position

$$X(t) = 3(\cos t, \sin t) + (\cos 15t, \sin 15t), \qquad t \in \left[0, \frac{\pi}{6}\right].$$

Determine the set of times t at which the polar angle of X with respect to the origin is changing clockwise.

2023.1.4. Let

$$S = \{2^k \mid k \text{ is a nonnegative integer}\} = \{1, 2, 4, 8, \dots\}.$$

For a positive integer n, let $f(n)$ be the number of ways to express n as a sum of elements of S, with each element used at most three times. For example,

$$7 = 1+1+1+4 = 1+1+1+2+2 = 1+2+2+2 = 1+2+4,$$

so $f(7) = 4$. Express $f(n)$ as a function of n.

2023.1.5.

(a) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real matrices. Prove that every eigenvalue of $\mathbf{AB}$ is an eigenvalue of $\mathbf{BA}$.

(b) Let n be an odd positive integer, and suppose that $n \times n$ real matrices $\mathbf{A}$ and $\mathbf{B}$ satisfy

$$2\mathbf{AB} = (\mathbf{BA})^2 + \mathbf{I}_n.$$

Prove that

$$\det(\mathbf{I}_n - \mathbf{AB}) = 0.$$

2023.1.6. Let

$$S = \{k \in \mathbb{Z} \mid 1 \le k \le 2023,\ \gcd(k, 2023) = 1\},$$

and define

$$N = \sum_{a \in S} \sum_{b \in S} \left\{ \frac{ab}{2023} \right\},$$

where $\{t\} = t - \lfloor t \rfloor$. Prove that N is a positive integer and find the prime factorization of N.

2024.0.1. Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix}.$$

Find real numbers a, b, c such that

$$\mathbf{A}^5 = a\mathbf{A}^2 + b\mathbf{A} + c\mathbf{I}.$$

2024.0.2. Find the real number N satisfying

$$\sum_{k=1}^{2024} (2025 - k)\, 2024^k\, 2025^{2024-k} = 2024^N.$$

2024.0.3. Let D be a bounded closed convex region in the coordinate plane, and let β be the boundary of D. Suppose that:

(i) β is a smooth simple closed curve.

(ii) For every point O on β, the circle of radius 1 centered at O meets β at exactly two points.

(iii) For every point O on β, the tangent line to β at O does not meet β at any point other than O.

Let M be the midpoint of a rod PQ of length 1. Move the rod through one full turn while keeping both endpoints P, Q on β at all times; that is, move P and Q continuously once around β. Suppose that the locus of M is a simple closed curve γ. Prove that the area of the region between β and γ is

$$\frac{\pi}{4}.$$

2024.2.1. Solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x + y - 2}{x - y}, \qquad y(1) = 0.$$

2024.2.2. For $a > 0$, define the solid

$$V = \left\{ (x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + (z-1)^2 \le 1, \quad z \ge \sqrt{ax^2 + ay^2} \right\}.$$

If the volume of V is

$$\frac{7\pi}{12},$$

find a.

2024.2.3. Evaluate the improper integral

$$\int_0^\infty \frac{1}{(x^{2024}+1)(x^2+1)}\,\mathrm{d}x.$$

2024.2.4. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real matrices, where $n \geq 2$, and suppose that

$$\mathrm{rk}(\mathbf{B}) = 1.$$

Prove that

$$\det(\mathbf{A}+\mathbf{B})\det(\mathbf{A}-\mathbf{B}) \leq \det(\mathbf{A})^2.$$

2024.1.1. Let S be the surface in $\mathbb{R}^3$ defined by

$$4z^2 = x^2 + y^2 - 1.$$

Prove that S contains exactly two lines passing through $P = (1, 2, 1)$. If θ is the acute angle between these two lines, find $\cos\theta$.

2024.1.2. Find the tens digit and the units digit of

$$37^{2024}.$$

2024.1.3. Let $f\colon \mathbb{R} \to \mathbb{R}$ be twice differentiable, with f'' continuous. Suppose that for every real x,

$$f'(x) > 0 \qquad \text{and} \qquad f''(x) < 0.$$

Prove that for every $t > 0$,

$$\left|\int_0^t \cos(f(x))\,\mathrm{d}x\right| \leq \frac{2}{f'(t)}.$$

2024.1.4. Let $n \geq 3$, and let real numbers $a_1, a_2, \ldots, a_n$ satisfy

$$a_1 + a_2 + \cdots + a_n = 0, \qquad a_1^2 + a_2^2 + \cdots + a_n^2 = 1.$$

Find the maximum value of

$$\left(\sum_{k=1}^{n-1} a_k a_{k+1}\right) + a_n a_1.$$

2025.0.1. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be differentiable and satisfy

$$f(0,0) = 0, \qquad \lim_{t\to 0}\frac{f(t,t)}{t} = 2, \qquad \lim_{t\to 0}\frac{f(t,2t)}{t} = -1.$$

For real numbers a, b satisfying

$$a^2 + b^2 = 1,$$

find the maximum and minimum possible values of

$$\lim_{t\to 0}\frac{f(at,bt)}{t}.$$

2025.0.2. Let n be a positive integer and let $a_{ij} > 0$ for $1 \le i, j \le n$. Define

$$A_n = \left(\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} \right)^2,$$

$$B_n = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} \left(\sum_{k=1}^{n} a_{ik} \right)^{1/2} \left(\sum_{k=1}^{n} a_{kj} \right)^{1/2},$$

and

$$C_n = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}^2.$$

Prove that

$$A_n \le nB_n \le n^2 C_n.$$

2025.2.1. Solve the differential equation

$$f''(x) + 2f'(x) + f(x) = 2x^{-2}e^{-x}, \qquad f(1) = f'(1) = 0.$$

2025.2.2. Five points A, B, C, D, E lie in a plane in three-dimensional vector space and satisfy the following conditions:

(i)
$$\|\overrightarrow{AB}\| = \|\overrightarrow{BC}\| = \|\overrightarrow{CD}\| = \|\overrightarrow{DE}\| = \|\overrightarrow{EA}\| = 1.$$

(ii)
$$\overrightarrow{AB} \times \overrightarrow{BC} = \overrightarrow{BC} \times \overrightarrow{CD} = \overrightarrow{CD} \times \overrightarrow{DE} = \overrightarrow{DE} \times \overrightarrow{EA} = \overrightarrow{EA} \times \overrightarrow{AB} \neq \mathbf{0}.$$

(iii)
$$\overrightarrow{AB} \cdot \overrightarrow{BC} = \overrightarrow{BC} \cdot \overrightarrow{CD} = \overrightarrow{CD} \cdot \overrightarrow{DE} = \overrightarrow{DE} \cdot \overrightarrow{EA} = \overrightarrow{EA} \cdot \overrightarrow{AB}.$$

(a) Prove that $\overrightarrow{CE}$ is a scalar multiple of $\overrightarrow{AB}$.

(b) Determine all possible values of k such that

$$\overrightarrow{CE} = k\overrightarrow{AB},$$

and describe the figure $ABCDE$ in each case.

2025.2.3. Let n be a positive integer and let

$$\alpha = \frac{\pi}{n+1}.$$

Define the $n \times n$ matrix

$$\mathbf{A} = \begin{pmatrix} \sin\alpha & \sin(2\alpha) & \cdots & \sin(n\alpha) \\ \sin(2\alpha) & \sin(4\alpha) & \cdots & \sin(2n\alpha) \\ \vdots & \vdots & \ddots & \vdots \\ \sin(n\alpha) & \sin(2n\alpha) & \cdots & \sin(n^2\alpha) \end{pmatrix}.$$

Find $|\det(\mathbf{A})|$.

2025.2.4. Let $\mathbf{A} = (a_{jk})$ be an $(n+1) \times (n+1)$ real matrix and let $\mathbf{B} = (b_{jk})$ be an $n \times n$ matrix whose entries are polynomials in the n variables $x_1, \ldots, x_n$. Define

$$a_{jk} = j^{(n+1-k)^2}, \qquad b_{jk} = b_{jk}(x_1, \ldots, x_n) = x_j^{(n+1-k)^2-1}.$$

Prove that

$$\det(\mathbf{A}) = (-1)^n (n!)^2 \int_n^{n+1} \cdots \int_2^3 \int_1^2 \det(\mathbf{B}) \, \mathrm{d}x_1 \, \mathrm{d}x_2 \cdots \mathrm{d}x_n.$$

2025.2.5. Let $n, k \geq 2$ be positive integers, and define

$$F \colon (0,1)^n \to \mathbb{R}$$

by

$$F(x_1, \ldots, x_n) = \int_{-\infty}^{\infty} \frac{1}{|t|^k + |t - x_1|^k + \cdots + |t - x_n|^k} \, \mathrm{d}t.$$

Find the range of positive real numbers p for which

$$\int_0^1 \cdots \int_0^1 F(x_1, \ldots, x_n)^p \, \mathrm{d}x_1 \cdots \mathrm{d}x_n$$

converges.

2025.1.1. Find the least positive integer k satisfying the following condition: for any k distinct integers, there always exist two distinct integers m, n among them such that

$$7 \mid (m^2 + 13n^2).$$

2025.1.2. For 2×2 real matrices $\mathbf{A}$ and $\mathbf{B}$, define

$$U(\mathbf{A}, \mathbf{B}) = \mathbf{ABAB} - \mathbf{BABA}.$$

Find all real numbers a for which there exist 2×2 real matrices $\mathbf{X}$ and $\mathbf{Y}$ satisfying

$$U(\mathbf{X}, \mathbf{Y})^{2025} - U(\mathbf{X}, \mathbf{Y})^{1101} = a\mathbf{I}.$$

2025.1.3. For positive integers n, k, define the point

$$P_{n,k} = \left(n \cos \frac{2k\pi}{n}, n \sin \frac{2k\pi}{n} \right).$$

Let $D_{n,k}$ be the interior of the circle centered at $P_{n,k}$ with radius $n+1$, and define

$$D_n = \bigcap_{k=1}^{n} D_{n,k}.$$

Evaluate

$$\lim_{n \to \infty} \iint_{D_n} x^2 e^{-(x^2+y^2)} \, \mathrm{d}x \, \mathrm{d}y.$$

2025.1.4. Let n be a positive integer and $\alpha \in \mathbb{R}$. Define the $n \times n$ matrix

$$\mathbf{A} = \begin{pmatrix} \sin\alpha & \sin(2\alpha) & \cdots & \sin(n\alpha) \\ \sin(2\alpha) & \sin(4\alpha) & \cdots & \sin(2n\alpha) \\ \vdots & \vdots & \ddots & \vdots \\ \sin(n\alpha) & \sin(2n\alpha) & \cdots & \sin(n^2\alpha) \end{pmatrix}.$$

For even n and

$$\alpha = \frac{\pi}{n+1},$$

find $\det(\mathbf{A})$.

2025.1.5. For a positive integer n, let a_n be the number of permutations π of $\{1, 2, \ldots, n\}$ satisfying both of the following conditions:

(i) For every integer i with $1 \le i \le n$,

$$(\pi \circ \pi)(i) = i.$$

(ii) There do not exist integers a, b, c, d such that

$$1 \le a < b < c < d \le n$$

and

$$\pi(a) > \pi(b) > \pi(c) > \pi(d).$$

Evaluate

$$\lim_{n\to\infty} a_n^{1/n}.$$